

Chapter 11 :

Frequency Response of Amplifiers

11.1 Introduction

In Chapter 6 and 9 we have discussed the AC analysis of BJT amplifiers and FET amplifiers. In those analyses, we have considered all external capacitors short, and all internal parasitic capacitances open. For an amplifier this will happen for a particular range of frequencies called **mid-band frequencies**. Thus, we have studied the response of BJT and FET amplifiers in mid-band frequencies. In this chapter, we will study the response of amplifiers to the frequencies where the external capacitors affect the performance. The range of frequencies at which the external capacitors cannot be considered short, are the lower frequencies than the mid-band frequencies. This response is called the **low frequency response**. We will also study the effect of BJT's and FET's internal parasitic capacitances on the performance of amplifiers. At some frequencies, higher than the mid-band frequencies, the parasitic capacitances start to reduce the performance of the amplifier. Response of the amplifiers to the frequencies for which the parasitic capacitances become effective is called **high frequency response**. In this chapter, we will study the low frequency response and high frequency response of different BJT and FET amplifiers. Before explaining the frequency response of amplifiers, we will discuss about dB unit and logarithmic scale, because dB unit and logarithmic plots are used for frequency response.

11.2 Decibel Unit

In most of the cases, we represent the gain (voltage gain, current gain, and power gain) of an amplifier by the ratio of the output to the input. This represents how big or small the output quantity is compared to the input quantity. However, in some cases, instead of representing the gain by this absolute ratio, it is represented in **decibel (dB)** unit. For example, to represent the losses in communication, in the frequency response of an amplifier the decibel unit is used. **Bel** is a logarithmic ($\log_{10}$) representation of the ratio of two quantities mainly two power levels. It is defined as,

$$Bel = \log_{10} \left(\frac{P_{mes}}{P_{ref}} \right) = \log_{10} \left(\frac{P_2}{P_1} \right) \quad (11-1)$$

Here, P_{mes} (or P_1) is the measured power, we are interested of and P_{ref} (or P_1) is the reference power. It was named the **Bel**, in honor of the telecommunications pioneer **Alexander Graham Bell**. Initially it was introduced to compare the sound levels. Being a very large unit (1 Bel means P_{mea} is 10 times larger than P_{ref}), Bel is seldom used. Because one decibel difference in loudness between two sounds is the smallest difference detectable by human hearing, the decibel was proposed as the working unit. In 1928, the authority of the **Bell Telephone Laboratories** renamed the **Transmission Unit** (used to represent the losses in telephone line) into the decibel. **Decibel**, denoted as dB, is defined as the one-tenth of a Bell, as

$$\text{dB} = 10 \log_{10} \left(\frac{P_{\text{mes}}}{P_{\text{ref}}} \right) = 10 \log_{10} \left(\frac{P_2}{P_1} \right) \quad (11-2)$$

The two power levels, P_1 and P_2 , are often the input and output power of a system, respectively. In that case, Equ.(11-2) represents the power gain of the system in decibels. If $P_2 > P_1$, Equ.(11-2) gives a positive number, and if $P_2 < P_1$, the equation gives a negative number. Thus, a positive dB implies gain of the system, while the negative dB implies attenuation. If $P_2 = P_1$, the result is 0 dB, since $\log_{10}(1) = 0$. So, 0 dB means that there is no change in the output power.

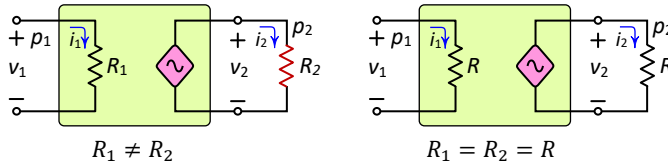


Fig.11-1: Inputs and outputs of amplifiers

Let R_1 be the resistance across which v_1 voltage is applied and P_1 power is developed. Similarly, let R_2 be the resistance across which v_2 voltage is applied and P_2 power is developed. Then, since $P = v^2/R$, from Equ.(11-2), we have

$$\text{dB} = 10 \log_{10} \left(\frac{P_2}{P_1} \right) = 10 \log_{10} \left(\frac{v_2^2/R_2}{v_1^2/R_1} \right)$$

or,

$$\text{dB} = 10 \log_{10} \left(\frac{v_2^2 R_1}{v_1^2 R_2} \right) \quad (11-3)$$

If the resistance values are the same at the two points, where the power comparison is made ($R_1 = R_2 = R$), then Equ.(11-3) becomes

$$\text{dB} = 10 \log_{10} \left(\frac{v_2^2 R}{v_1^2 R} \right) = 10 \log_{10} \left(\frac{v_2}{v_1} \right)^2 = 20 \log_{10} \left(\frac{v_2}{v_1} \right) \quad (11-4)$$

Equ.(11-4) gives power gain (or loss) in terms of the voltage levels at two points of a circuit. To use this equation for power comparison the resistances across which the powers are developed must be equal. The same equation is used to compare voltage levels regardless of the resistance values of the two points. In other word, practically we can use Equ.(11-4) to calculate the voltage gain of amplifier as,

$$\text{dB (Voltage gain)} = 20 \log_{10} \left(\frac{v_2}{v_1} \right) \quad (11-5)$$

If the resistances, across which the voltages are applied, are equal, then the voltage gain in dB will be equal to the power gain in dB, as found in Example 11-1.

Example 11-1

Both the input impedance and the output impedance of an amplifier are equal and is $R_i = R_o = 1 \text{ k}$. If 2 V rms is applied at the input, 10 V rms is developed at the output. Calculate the power gain using Equ.(11-2) and the voltage gain using Equ.(11-5).

Solution:

Power at the input will be,

$$P_i = \frac{v_i^2}{R_i} = \frac{(2 \text{ V})^2}{1 \times 10^3 \Omega} = 4 \times 10^{-3} \text{ W}$$

Power at the output will be,

$$P_o = \frac{v_o^2}{R_o} = \frac{(10 \text{ V})^2}{1 \times 10^3 \Omega} = 100 \times 10^{-3} \text{ W}$$

Using Equ.(11-2),
$$A_p = 10 \log_{10} \left(\frac{P_o}{P_i} \right) = 10 \log_{10} \left(\frac{100 \times 10^{-3} \text{ W}}{4 \times 10^{-3} \text{ W}} \right)$$

or,
$$A_p = 10 \log_{10}(25) \approx 13.8 \text{ [Ans.]}$$

The voltage gain of the amplifier can be calculated as,

Using Equ.(11-5),
$$A_v = 20 \log_{10} \left(\frac{v_o}{v_i} \right) = 20 \log_{10} \left(\frac{10 \text{ V}}{2 \text{ V}} \right)$$

or,
$$A_v = 20 \log_{10}(5) \approx 13.8 \text{ [Ans.]}$$

Comments: As the resistances at two points are the same, the power gain and the voltage gain are also same.

It will be very helpful to understand the frequency response of different amplifiers, if some common dB values are known. A two-to-one ($A_v = 2$) change in voltage corresponds to approximately 6 dB, a eight-to-one change ($A_v = 8$) approximately corresponds to 18 dB. Similarly, a ten-to-one voltage change corresponds to approximately 20 dB. The dB value will be positive, if the voltage increases and will be negative if the voltage decreases. To calculate these values, let us consider that the input voltage of an amplifier is $v_i = 1 \text{ V}$. If this voltage at the output of the amplifier is doubled ($v_o = 2 \text{ V}$), then the absolute voltage gain of the amplifier will be, $A_v = 2$. It can be calculated in dB scale as,

$$A_v(\text{dB}) = 20 \log_{10} \left(\frac{v_o}{v_i} \right) = 20 \log_{10} \left(\frac{2 \text{ V}}{1 \text{ V}} \right) \approx 6 \text{ dB}$$

But, if the output voltage is half of the input ($v_o = 0.5 \text{ V}$), then

$$A_v(\text{dB}) = 20 \log_{10} \left(\frac{v_o}{v_i} \right) = 20 \log_{10} \left(\frac{0.5 \text{ V}}{1 \text{ V}} \right) \approx -6 \text{ dB}$$

If the output voltage is eight times the input voltage ($v_o = 8 \text{ V}$), then

$$A_v(\text{dB}) = 20 \log_{10} \left(\frac{v_o}{v_i} \right) = 20 \log_{10} \left(\frac{8 \text{ V}}{1 \text{ V}} \right) \approx 18 \text{ dB}$$

Again, if we consider the output voltage is ten times the input voltage ($v_o = 10 \text{ V}$, and $A_v = 10$) then the gain of the amplifier in dB will be,

$$A_v(\text{dB}) = 20 \log_{10} \left(\frac{v_o}{v_{in}} \right) = 20 \log_{10} \left(\frac{10 \text{ V}}{1 \text{ V}} \right) \approx 20 \text{ dB}$$

Similarly, for $v_o = 1/8 \text{ V}$, and $v_o = 1/10 \text{ V}$, the gain of the amplifier will be -18 dB , and -20 dB , respectively.

Use of dB Unit for Cascaded (Multi-Stage) Amplifier:

In a cascaded amplifiers system (multi-stage amplifier), the overall gain of the system will be the algebraic sum of the gains in dB values of individual amplifiers. If the voltage is doubled, an additional 6 dB is added to the voltage gain, and every time it is increased by a factor of 10, an additional 20 dB is added to the voltage gain. On the other hand, if the voltage is halved at any stage, 6 dB will be subtracted from the voltage gain. We can calculate the dB value of a voltage gain by breaking it into values of known dB. For example, the voltage gain of $4 = 2 \times 2$ corresponds to $(6 \text{ dB} + 6 \text{ dB})$ 12 dB, and the voltage gain of $80 = 8 \times 10$ corresponds to $(18 \text{ dB} + 20 \text{ dB})$ 38 dB. It is further illustrated in Example 11-2

Example 11-2

Fig.11-2 shows a two-stage cascaded amplifier. The input voltage of the first amplifier is $v_i = 2 \text{ mV}$ (rms). If the gain of the first amplifier is 27 dB, calculate the output voltage of this stage. If the voltage at the output of the 2nd amplifier is 0.5 V (rms), then calculate the voltage gain of this stage. Also calculate the overall gain of the whole system.

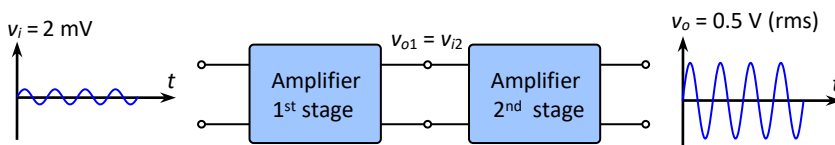


Fig.11-2: Two-stage cascaded amplifier for Example 10.2

Solution:

Let the output voltage of the first stage be, v_{o1}

$$\therefore A_{v1}(\text{dB}) = 27 = 20 \log_{10} \left(\frac{v_{o1}}{v_i} \right) = 20 \log_{10} \left(\frac{v_{o1}}{2 \times 10^{-3}} \right)$$

$$\text{or,} \quad \log_{10} \left(\frac{v_{o1}}{2 \times 10^{-3} \text{ V}} \right) = \frac{27}{20} = 1.35$$

$$\text{or,} \quad \frac{v_{o1}}{2 \times 10^{-3} \text{ V}} = 10^{1.35} = 22.387$$

$$\therefore v_{o1} = 22.387 \times 2 \times 10^{-3} \text{ V} = 44.77 \text{ mV [Ans.]}$$

The input voltage of the 2nd stage will be the output voltage of the 1st stage. So the gain of the 2nd stage will be,

$$A_{v2}(\text{dB}) = 20 \log_{10} \left(\frac{v_{o2}}{v_{i2}} \right) = 20 \log_{10} \left(\frac{0.5 \text{ V}}{44.77 \times 10^{-3}} \right) \approx 20.96 \text{ dB [Ans.]}$$

Overall gain of the cascaded amplifier will be,

$$A_v = 20 \log_{10} \left(\frac{v_{o2}}{v_i} \right) = 20 \log_{10} \left(\frac{0.5 \text{ V}}{2 \times 10^{-3}} \right) \approx 47.96 \text{ dB [Ans.]}$$

Comments: The overall voltage gain of the cascaded system is the sum of individual voltage gains in dB. It is clear from this example (27 dB + 20.96 dB = 47.96 dB).

11.3 dBm and Other Units

We have seen that decibels are derived from the ratio of two powers or voltages. Therefore, it represents a comparison of one voltage or power level to another. It is correct to speak of power gain or voltage gain in terms of decibels but, it is meaningless to speak of the output level in dB, unless the reference level is specified. To overcome this problem, in some applications a standard reference level is used for all dB computation. One such reference is 1 mW. The resistance associated with the 1 mW power level is 600 Ω chosen, because it is the *characteristic impedance* of audio transmission lines. When the reference is 1 mW, the unit is called *decibel milliwatt* and is represented as dBm and is given by,

$$\text{dBm} = 10 \log_{10} \left(\frac{P_{\text{mes}}}{P_{\text{ref}}} \right) = 10 \log_{10} \left(\frac{P_{\text{mes}}}{1 \text{ mW}} \right) \quad (11-6)$$

In this case 0 dBm corresponds to 1 mW. Similarly, 3 dBm and 6 dBm corresponds to 2 mW and 4 mW, respectively. That is, 1 mW will be multiplied by 2 for an increase of 3 dBm and so on.

Another standard reference is 1 W. When 1 W is used as the reference power, the unit is called *decibel watt* and is represented as,

$$\text{dBW} = 10 \log_{10} \left(\frac{P_{\text{mes}}}{P_{\text{ref}}} \right) = 10 \log_{10} \left(\frac{P_{\text{mes}}}{1 \text{ W}} \right) \quad (11-7)$$

Here also, 0 dBW corresponds to 1 W and 3 dBm corresponds to 2 W.

Sometimes, to represent the voltage level at different points of a system, 1 V reference is used. Then the unit is called *decibel volt* (dBV) and is defined as,

$$\text{dBV} = 20 \log_{10} \left(\frac{V_{\text{mes}}}{V_{\text{ref}}} \right) = 20 \log_{10} \left(\frac{V_{\text{mes}}}{1 \text{ V}} \right) \quad (11-8)$$

11.4 Logarithmic Plot

It is a convenient way and widely followed practice to plot the logarithm of frequency response data rather than actual values. If the logarithm of frequency is plotted along the x -axis, a wide frequency range can be displayed in a graph paper without losing resolution at the low frequency

end. For example, let us consider the passive band-pass filter of Fig.11-3. The response of this filter is shown in Fig.11-4(a) using the absolute frequency data, and in Fig.11-4(b) using the logarithmic data. In both cases y-axis represents the gain of the circuit in actual values. As shown in Fig.11-4(a), for linear spacing of frequency, it is not possible to find the lower cut-off frequency at all. However, to get the resolution, if we decrease the frequency values of the smallest unit, a very large graph paper will be needed or we will lose the data in the higher frequency region. On the other hand the logarithmic plot of Fig.11-4(b) gives the better resolution both on the low-frequency range and in the high-frequency range. In this graph, we can easily determine the lower and upper cut-off frequency. The graph that uses logarithmic data on one axis and actual data on the other axis is called the **semi-log graph**.

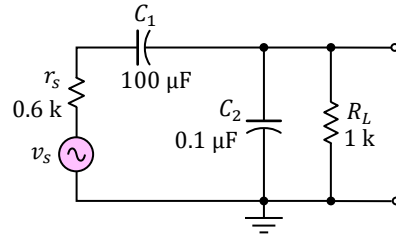


Fig.11-3: A passive band-pass filter

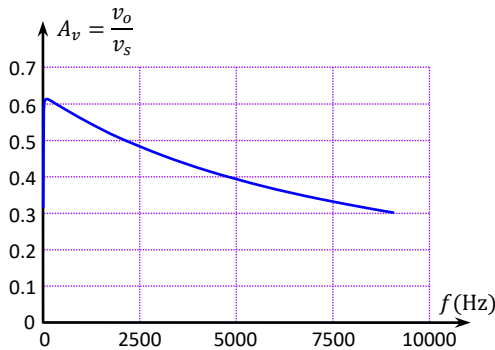


Fig.11-4(a): Response of the band-pass filter of Fig.11-3 using absolute scale

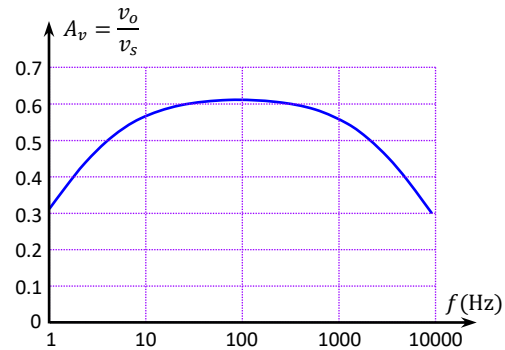


Fig.11-4(b): Response of the band-pass filter of Fig.11-3 using logarithmic scale

Fig.11-5 shows a semi-log graph paper with more details. The x -axis has logarithmic scale, and the y -axis has linear scale. Look at the x -axis, it has five repeated patterns which are called **cycles**. This graph paper has five cycles and in every cycle, there are nine smallest units. These cycles can be labeled according to the requirement for the data values on this axis. Here, we have labeled starting from 0.1 and up to 10^4 . Remember that there is no 0 point on the logarithmic scale, no matter how many cycles are present in the graph, because $\log_{10} 0 = -\infty$. In the logarithmic scale, any one-to-ten range of values is called a **decade**. For example, 10 to 100 is a decade, and 100 to 1000 is another decade. There are five decades in this graph paper. The range of data within a decade is divided by 9 and is represented by the smallest units within them. For example, a smallest unit of 0.1 to 1.0 decade will be 0.1, while a smallest unit of 10^2 to 10^3 decade will be 100. An octave is any one-to-two range of values, such as 10 to 20, 100 to 200, 200 to 400 etc. Look at the figure, like the decades, every octave occupies the same distance on the graph paper. Generally, the frequency response and phase-shift of an amplifier are presented in semi-log papers. Graphs of frequency response, when plotted against the logarithm of frequency, are called **Bode plots**. If the gain of an amplifier does not vary so much or if the gain is calculated in dB, then we can use semi-log graph. If the variation of gain is very large, and calculated in actual value (not dB), then we have to use log-log graph.

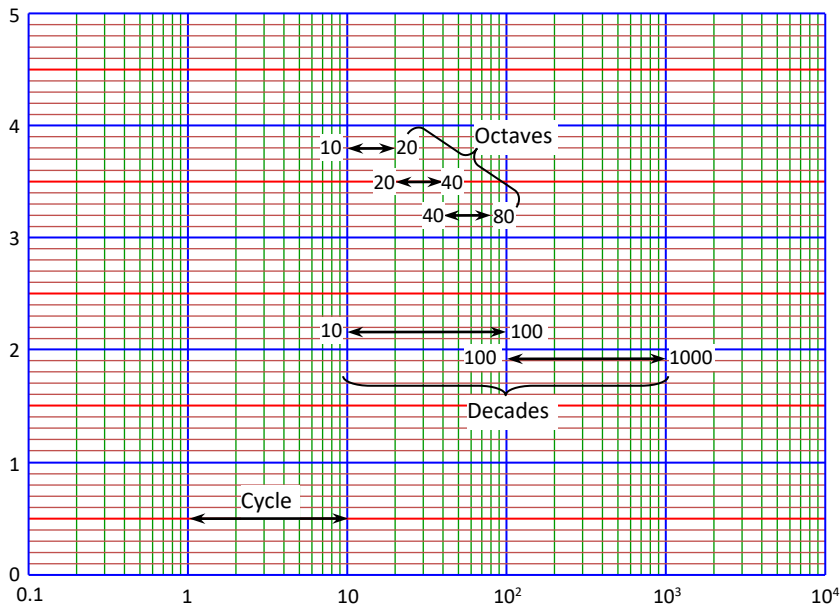


Fig.11-5: Semi-log graph paper

If a graph paper has logarithmic scales on both the axis, then it is called **log-log** graph paper. For example, a two-cycle-by-five-cycle (2×5) log-log graph paper is given in Fig.11-6. Moreover, this figure shows the effect of using logarithmic scale for gain.

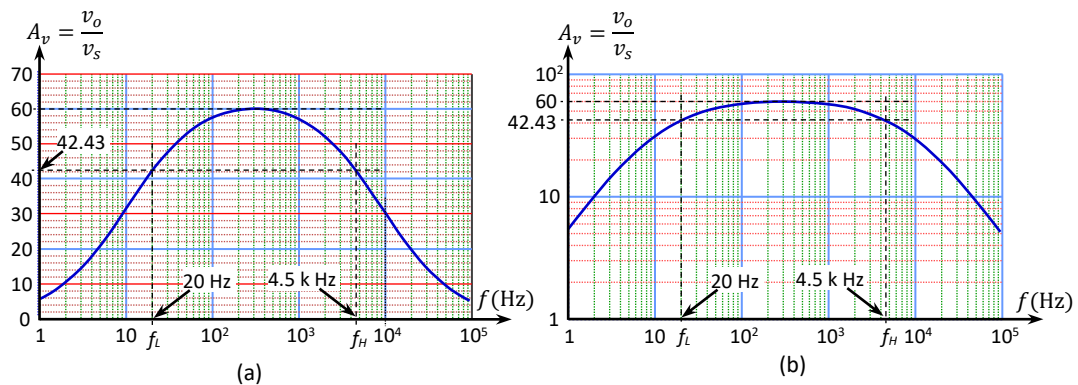


Fig.11-6: Frequency response of an amplifier: (a) Semi-log plot, and (b) Log-log plot

Example 11-3

Fig.11-6(a) and Fig.11-6(b) show the frequency response of a typical amplifier in semi-log and log-log graph, respectively. Determine the lower and higher cut-off frequency of the amplifier from these graphs.

Solution:

From both the graph, we find that the maximum gain of the amplifier is 60. Therefore, at the lower and higher cut-off frequencies, the value of the gain will be,

$$A_{v(\text{cut-off})} = \frac{A_{v(\text{max})}}{\sqrt{2}} = \frac{60}{\sqrt{2}} \approx 42.43$$

In both the graphs, a horizontal line drawn along $A_v = 42.43$ has been drawn. This horizontal line intersects the graphs at two points. The first intersecting point gives the lower cut-off frequency value which is 20 Hz. The second intersecting point gives the higher cut-off frequency value which is 4.5 kHz.

Answer: $f_L = 20$ Hz, and $f_H \approx 4.5$ kHz are shown in the graphs of Fig.11-3.

Comments: Both the graphs give the same results.

11.5 Some Related Topics

In going to the frequency response of different BJT and FET amplifiers, we will discuss some related terms, to understand them. The terms are described in the following sub-headings.

11.5.1 Frequency Response

The **frequency response** of any electronic device or system is the variation, if any, in the level of its output signal when the frequency of the input signal is changed. In other words, **frequency response** is the manner in which the device or the system responds to change of frequency in the

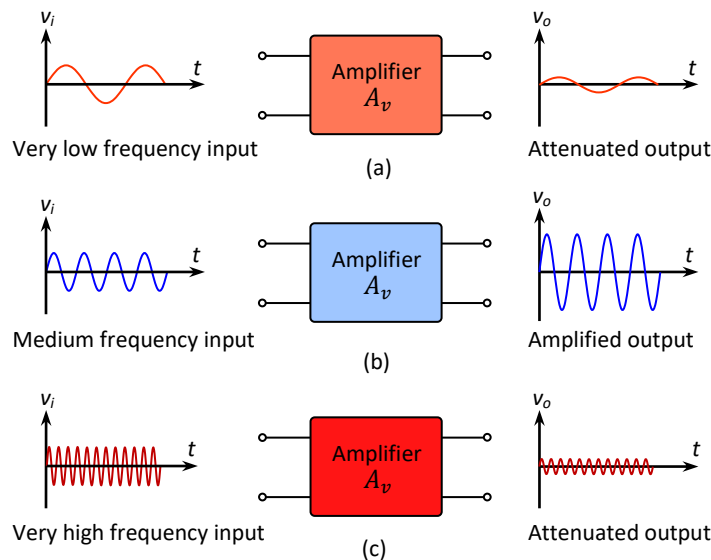


Fig.11-7: Response of an amplifier to different frequency-inputs

input signal. For example, the output voltage and, hence, the gain of an amplifier decreases for very low and high input-signal frequency, even, if the amplitude of the input remains same. This situation is illustrated in Fig.11-7. In this figure, the amplitude of the input signals is same for three cases, but the frequency has been changed. In the first case [Fig.11-7(a)], the frequency of the input signal is very low and the amplitude of the output is lower than that of input signal. It means

that, instead of being amplified, the input signal has been attenuated at the output, due to its very low frequency. Similarly, the input signal has been attenuated at the output due to very high frequency input is shown in Fig.11-7(c). On the other hand, in Fig.11-7(b) the input signal has medium frequency and, therefore, has been properly amplified at the output.

Although here the cases of three discrete frequencies have been explained, the gain of the amplifier increases gradually with the increase in frequency. At the medium frequency range the amplifier gives the maximum gain. Again for higher frequencies, the gain of the amplifier falls gradually. This situation is illustrated in Fig.11-8. The graph shows the change of gain of the amplifier with the change in frequency. This graph is called the frequency response of an amplifier.

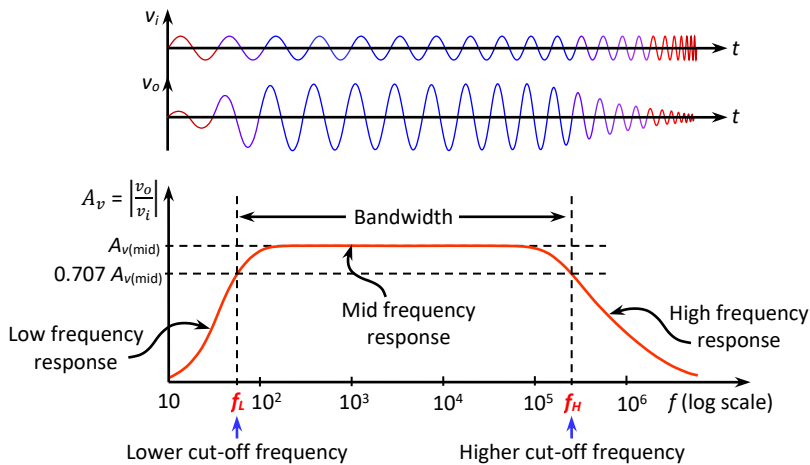


Fig.11-8: Variation of voltage gain of an amplifier with the variation of input frequency

11.5.2 Phase Response

The **phase response** of any electronic circuit or system is the variation of phase angle of its output signal when the frequency of the input signal is changed. Variation of the output level (amplitude, or RMS value of voltage, or power) of the output signal is usually accomplished by a variation in the phase angle of the output relative to the input signal. In fact it is the Phase-vs-frequency graph. If the input signal is $v_i = V_i \sin(\omega t + \phi_i)$ and the output signal is $v_o = V_o \sin(\omega t + \phi_o)$, the phase difference, or simply the phase angle, can be calculated as,

$$\phi = \phi_o - \phi_i \quad (11-9)$$

11.5.3 Bode Plot

The **Bode plot** is a graphical response prediction technique that is mainly used for amplifiers, filters, transmission line, and control circuits. It is named after **Hendrik Wade Bode** (boodi/boh-dee, period: 1906-1982), an American engineer known for his work in control systems theory and telecommunications. A **Bode plot** is actually a pair of plots: One graph shows the gain of a system versus frequency, while the other graph shows the circuit phase versus frequency. Bode plots use semi-log graphs with the gain always in dB scale. Fig.11-9 and Fig.11-10 show the Bode plots of a typical amplifier. Fig.11-9 is the gain response, while Fig.11-10 is the phase response.

Note how the plot is relatively flat in mid-band region (Fig.11-9). The gain value in this region is known as the **mid-band gain**. At either extreme of the mid-band region, the gain begins to decrease. The gain plot shows two important frequencies, f_L and f_H . f_L is the lower break frequency (also called **lower cut-off frequency**) while f_H is the upper break frequency (also called **higher cut-off frequency**).

The gain at the break frequencies is 3 dB less than the mid-band gain. These frequencies are also known as the **half-power points**, or **corner frequencies**. Normally, amplifiers are only used to amplify signals between f_L and f_H . The break points are caused by the presence of circuit reactances, typically coupling and stray capacitances. The gain plot is a summation of the mid-band response with the upper and lower frequency limiting networks. It is possible to design amplifiers with no lower break frequency (i.e., a DC amplifier), however, all amplifiers will exhibit an upper cut-off frequency.

The exact shape of the roll-off (gain decreasing) regions will depend on the design of the circuit. The roll-off in the lower and higher cut-off frequency regions depends on the number of cut-off frequencies and their positions.

If the amplifier has only one lower cut-off frequency, the roll-off rate in the lower frequency side is 20 dB/decade (or 6 dB/octave). But if the amplifier has more than one, say n number of lower cut-off frequencies of same value, the roll-off rate will be $n \times 20$ dB/decade (or $n \times 6$ dB/octave). But, if n number of lower cut-off frequencies are far apart from one another, n number of break points will be found in the roll-off regions. The same roll-off rate will occur in the high frequency region as shown in Fig.11-9.

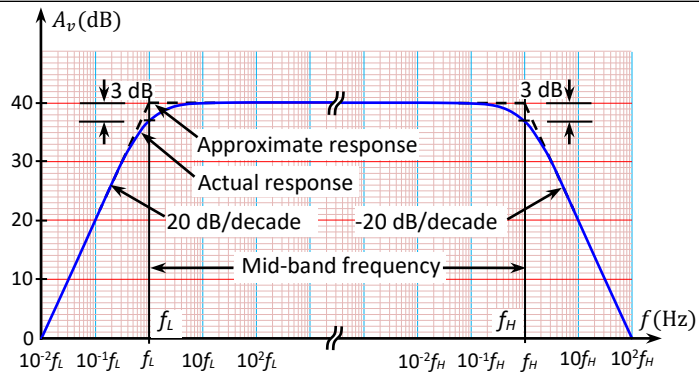


Fig.11-9: Bode plot: Gain response

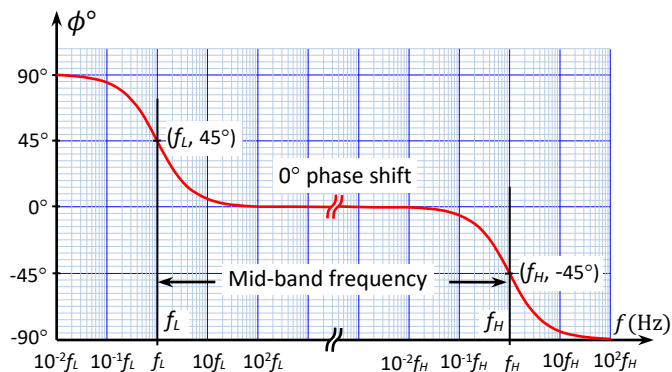


Fig.11-10: Bode plot: Phase response

Fig.11-10 shows the variation of phase shift of the output signal considering the phase of the input signal as 0° and the amplifier has 0° phase shift (like CB and CC amplifiers). At very low frequency, like $0.01 \times f_L$, the output will lead the reference input signal by 90° . Then with the increase in frequency this phase shift decreases with frequency. At lower cut-off frequency the phase shift is 45° . At $100 \times f_L$ frequency, the phase shift becomes almost 0° , that is there is no phase shift of output from input. This phase shift is produced by the coupling (and bypass) capacitors of the

amplifier that works in series (or parallel). In fact, these are the external capacitors used in the circuit not the parasitic capacitances of the device.

In the high frequency regions, the output lags the input signal. At the higher cut-off frequency ($f = f_H$) the phase difference between output and input is -45° , that is, the output lags the input by 45° . At very high frequency, like by $100 \times f_H$, the phase difference is almost -90° . At frequencies below $0.01 \times f_H$ the phase difference is practically 0° . This phase shift is produced by the parasitic capacitances that work in parallel to the input and output of the amplifiers.

From this discussion, we find that an amplifier can produce 45° to -45° phase shift of the output signal when it amplifies the signals having frequencies between lower and higher cut-off frequencies. Fig.11-10 shows the phase response of the amplifier that has only one lower cut-off and one higher cut-off frequency. But the phase response of the amplifier with more than one lower and higher cut-off frequencies will be different from this. The frequency range extending from f_L to f_H is the **bandwidth** of the amplifier.

11.6 Series Capacitance and Low-frequency Response

The lower cut-off frequency of an amplifier is imposed by the capacitors connected in series with the signal flow (true for even the bypass capacitor). The most common capacitors that come in series with the signal flow are the amplifier's input and output coupling capacitors (C_{ic} , and C_{oc}). Another very important capacitor that works in series with the signal flow is the bypass capacitor. In the AC analysis of amplifiers, we have considered these capacitors as short circuit. At low frequencies, the reactance of these capacitors becomes very large, so a significant portion of the AC signal is dropped across them and the amplifier's gain decreases. As frequency approaches 0 (DC voltage), the reactances of these capacitors approach infinity (open circuit) and hence the gain of the amplifier approaches 0. In this section, we will explore how large reactance affect the overall gain of the amplifier.

Fig.11-11 shows the simplified diagram of an amplifier where the input and output impedance of the amplifier are represented as Z_i and Z_o , respectively. The source signal (v_s) is coupled to the input of the amplifier by the coupling capacitor C_{ic} . Here, the internal resistance of the signal source has been considered zero. The effect of C_{ic} and C_{oc} has been described below:

11.6.1 Effect of Input Coupling Capacitor

In the circuit of Fig.11-11, note that the input signal will be divided between the reactance of the capacitor ($1/\omega C_{ic}$) and the input resistance of the amplifier (Z_i). The voltage across the input resistance will work as the input voltage for the amplifier and its value can be calculated using voltage divider rule as,

$$v_i = \frac{v_s \times Z_i}{Z_i - jX_{C_{ic}}} \quad (11-10)$$

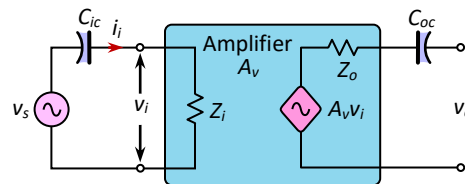


Fig.11-11: Two-port model of an amplifier

where,

$$X_{Cic} = \frac{1}{\omega C_{ic}} \Omega \quad (11-11)$$

and

$$\omega = 2\pi f \text{ rad/s} \quad (11-12)$$

From Equ.(11-10), we can calculate the magnitude of v_i as,

$$|v_i| = \frac{Z_i}{\sqrt{Z_i^2 + X_{Cic}^2}} |v_s| = \frac{Z_i}{\sqrt{Z_i^2 + (1/\omega C_{ic})^2}} |v_s| \quad (11-13)$$

or,

$$|v_i| = \frac{\omega Z_i C_{ic}}{\sqrt{1 + (\omega Z_i C_{ic})^2}} |v_s| \quad [\text{multiplying by } \omega C_{ic}] \quad (11-14)$$

From Equ.(11-14), we find that $|v_i| = 0$, when $\omega = 0$ (DC), and approaches $|v_s|$ as ω increases. At frequency $\omega = 1/Z_i C_{ic}$ rad/s, from Equ.(11-14) we find,

or,

$$|v_i| = \frac{(1/Z_i C_{ic}) Z_i C_{ic}}{\sqrt{1 + [(1/Z_i C_{ic}) Z_i C_{ic}]^2}} |v_s| \quad (11-15)$$

or,

$$|v_i| = \frac{1}{\sqrt{1+1}} |v_s| = \frac{|v_s|}{\sqrt{2}} \approx 0.707 |v_s| \quad (11-16)$$

For mid-band frequencies, the capacitor behaves as short circuit, so full v_s drops across the input resistance, and the amplifier gives the maximum voltage gain. However, when $\omega = 1/Z_i C_{ic}$, the amplifier input voltage falls to 0.707 times the source voltage, and hence the gain of the amplifier falls to 0.707 times the maximum gain. Therefore, we can say $\omega = 1/Z_i C_{ic}$ is the lower cut-off frequency of the amplifier. As the lower cut-off frequency is generally denoted as ω_L , we can write,

$$\omega_{Lic} = \frac{1}{Z_i C_{ic}} \text{ rad/s} \quad (11-17)$$

or,

$$f_{Lic} = \frac{1}{2\pi Z_i C_{ic}} \text{ Hz} \quad (11-18)$$

The condition to arise the lower cut-off frequency is,

$$\omega_L = \frac{1}{Z_i C_{ic}}$$

or,

$$Z_i = \frac{1}{\omega_L C_{ic}} = X_{Cic} \quad (11-19)$$

From Equ.(11-19), we can say that the lower cut-off frequency is the frequency at which the reactance of the series capacitor X_{Cic} equals the input impedance (resistance) Z_i .

So far we have neglected the source resistance. Now, if we consider the source resistance, r_s , Equ.(11-14) will be,

$$|v_i| = \frac{\omega Z_i C_{ic}}{\sqrt{1 + [\omega(r_s + Z_i) C_{ic}]^2}} |v_s| \quad (11-20)$$

And the lower cut-off frequency will be,

$$\omega_{Lic} = \frac{1}{(r_s + Z_i)C_{ic}} \text{ rad/s} \quad (11-21)$$

$$f_{Lic} = \frac{1}{2\pi(r_s + Z_i)C_{ic}} \text{ Hz} \quad (11-22)$$

The cut-off frequency defined by Equ.(11-21) and Equ.(11-22) is the frequency at which the ratio $|v_i|/|v_s|$ is 0.707 time the mid-band value, namely, 0.707 times $Z_i(r_s + Z_i)$. The ratio $|v_i|/|v_s|$ can be written in terms of the cut-off frequency f_L and the signal frequency f as,

$$\text{From Equ.(11-20),} \quad |v_i| = \frac{\omega Z_i C_{ic}}{\omega(r_s + Z_i)C_{ic}\sqrt{[1/\omega(r_s + Z_i)C_{ic}]^2 + 1}} |v_s| \quad (11-23)$$

$$\text{or,} \quad \frac{|v_i|}{|v_s|} = \frac{Z_i/(r_s + Z_i)}{\sqrt{[1/\omega(r_s + Z_i)C_{ic}]^2 + 1}}$$

Putting the value of ω_L from Equ.(11-21),

$$\frac{|v_i|}{|v_s|} = \frac{Z_i/(r_s + Z_i)}{\sqrt{[\omega_L/\omega]^2 + 1}} = \frac{K}{\sqrt{[\omega_L/\omega]^2 + 1}}$$

$$\text{or,} \quad \frac{|v_i|}{|v_s|} = \frac{K}{\sqrt{[2\pi f_L/2\pi f]^2 + 1}}$$

$$\therefore \quad \frac{|v_i|}{|v_s|} = \frac{K}{\sqrt{1 + \left(\frac{f_L}{f}\right)^2}} \quad (11-24)$$

$$\text{Where,} \quad K = \frac{Z_i}{r_s + Z_i} \quad (11-25)$$

To discuss about the phase response, let us consider the phase angle of the source signal as the reference phase, that is, let $\angle v_s = 0^\circ$. From Equ.(11-10), the phase angle of v_i will be,

$$\angle v_i = -\tan^{-1}\left(\frac{-X_{Cic}}{Z_i}\right) + \angle v_s = \tan^{-1}\left(\frac{X_{Cic}}{Z_i}\right) = \tan^{-1}\left(\frac{1}{\omega Z_i C_{ic}}\right) \quad (11-26)$$

Again, if we consider the source internal resistance (r_s), the phase angle of v_i will be,

$$\angle v_i = \tan^{-1}\left[\frac{1}{\omega(Z_i + r_s)C_{ic}}\right] \quad (11-27)$$

At the cut-off frequency, $\omega = \omega_L = 1/(Z_i + r_s)C_{ic}$. So from Equ.(11-22) we find,

$$\angle v_i = \tan^{-1}(1) = 45^\circ \quad (11-28)$$

It means that at cut-off frequency, the input voltage v_i leads v_s by 45° . To calculate the phase angle of v_i in terms of cut-off frequency and the signal frequency, we can rewrite Equ.(11-27) as,

$$\angle v_i = \tan^{-1}\left[\frac{1/(Z_i + r_s)C_{ic}}{\omega}\right]$$

$$\therefore \angle v_i = \tan^{-1} \left(\frac{\omega_L}{\omega} \right) \quad (11-29)$$

$$\text{or,} \quad \angle v_i = \tan^{-1} \left(\frac{2\pi f_L}{2\pi f} \right) = \tan^{-1} \left(\frac{f_L}{f} \right)$$

$$\therefore \angle v_i = \tan^{-1} \left(\frac{f_L}{f} \right) \quad (11-30)$$

From this equation, we can say, $\angle v_i$ increases with the decrease of f and vice versa.

Example 11-4

The amplifier shown in Fig.11-13 has mid-band gain, $v_o/v_s = 40$ dB. (i) Calculate the absolute mid-band gain and the gain at lower cut-off frequency both in dB and absolute value. (ii) Lower cut-off frequency, and phase shift of v_i at $f = 1$ kHz.

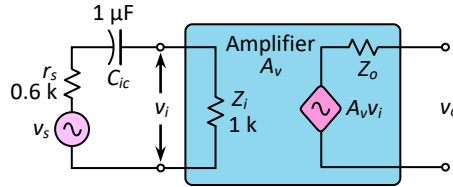


Fig.11-12: Amplifier with input coupling capacitor

Solution:

(i) Absolute mid-band gain,

$$40 \text{ dB} = 20 \log_{10} \left(\frac{v_o}{v_s} \right) \text{ dB} \quad \text{or,} \quad \frac{v_o}{v_s} = 10^{\left(\frac{40}{20}\right)} = 100 \text{ [Ans.]}$$

Gain at lower cut-off frequency,

$$A_{vL} = \frac{100}{\sqrt{2}} \approx 70.71 \text{ [Ans.]}$$

and

$$A_{vL}(\text{dB}) = 20 \log_{10} 70.71 \approx 37 \text{ dB [Ans.]}$$

(ii) Cut-off frequency,

$$\text{Using Equ.(11-22),} \quad f_L = \frac{1}{2\pi(r_s + Z_i)C_{ic}} = \frac{1}{2\pi(600 + 1000)1 \times 10^{-6}} \text{ Hz}$$

$$\therefore f_L \approx 99.5 \text{ Hz [Ans.]}$$

Phase shift of v_i w.r.t. v_s at $f = 500$ Hz,

$$\angle v_i = \tan^{-1} \left(\frac{f_L}{f} \right) = \tan^{-1} \left(\frac{99.5 \text{ Hz}}{500 \text{ Hz}} \right) \approx 11.26^\circ \text{ [Ans.]}$$

Comments: As the output voltage is measured with open circuit, Z_o will have no effect on the voltage gain.

To observe the frequency response graphically, the frequency responses of the amplifier of Example 11-4 are plotted in Fig.11-13. Fig.11-13(a) shows the variation of voltage gain in dB with frequency, in a semi-log graph. On the other hand, Fig.11-13(b) shows the variation of absolute gain (v_o/v_s) with frequency, in a log-log graph. The shapes of the graphs are exactly same.

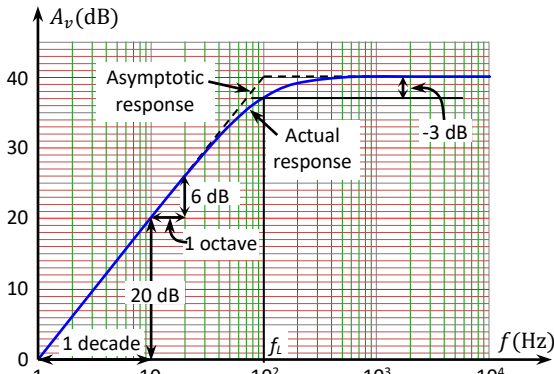


Fig.11-13(a): Frequency response in semi-log graph paper

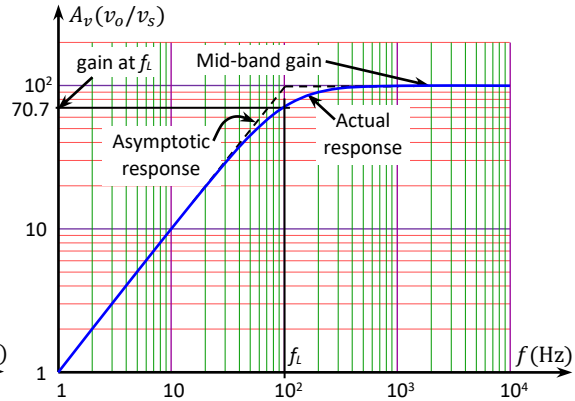


Fig.11-13(b): Frequency response in log-log graph paper

Fig.11-14 shows the variation of phase angle of v_i with respect to v_s in another semi-log graph. The voltage gain (A_v) and the lower cut-off frequency (f_L) of the amplifier is 100 (or 40 dB) and 99.5 Hz, respectively. As shown in Fig.11-13(a), at the cut-off frequency, the gain is 37 dB, that is 3 dB below the maximum gain. Similarly, in Fig.11-13(b), we find that at the cut-off frequency the gain is 70.7, that is 0.707 times the maximum gain. Moreover, from Fig.11-13(a), we find that the gain changes by 6 dB for every **octave** change in frequency, and changes by 20 dB for every **decade** change in frequency to the left side of cut-off frequency (roll-off region). This decrease of gain with frequency is called 20 dB/decade roll-off. One octave above the cut-off frequency the gain is 89.5% of mid-band gain, and one octave below, it is 44.7%. Again, one decade above f_L the gain is 99.5% of mid-band gain, and one decade below, it is 9.95%.

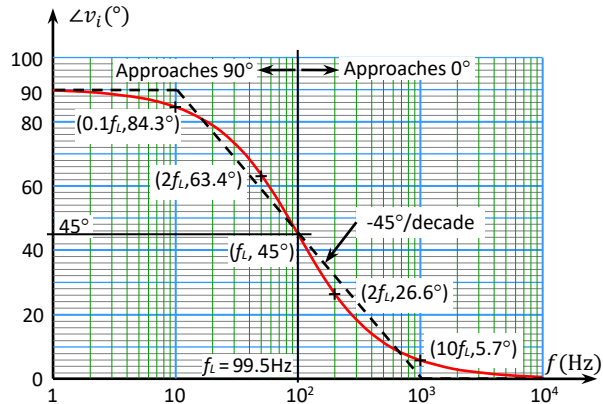


Fig.11-14: Phase response (Phase plot)

Fig.11-14 shows that the phase shift is 45° at the cut-off frequency. One octave above the cut-off frequency, the phase shift is 26.6° , and one octave below, it is 63.4° . Again, one decade above the cut-off frequency the phase shift is 5.7° , and one decade below, it is 84.3° . As shown in Fig.11-13(a) and Fig.11-13(b) the asymptotic response breaks at the point $f = f_L$. For this reason f_L is also called **break point frequency**. The slope of the asymptote is 6 dB/octave or 20 dB/decade.

One octave above the cut-off frequency the phase shift is 26.6° , and one octave below it is 63.4° . Again, one decade above the cut-off frequency the phase shift is 5.7° , and one decade below it is 84.3° .

11.6.2 Effect of Output Coupling Capacitor

The two-port model of the amplifier of Fig.11-11 is redrawn in Fig.11-15 to discuss the effect of output coupling capacitor. Now the load resistance is connected to the amplifier through output coupling capacitor C_{oc} . Here, the output resistance of the amplifier (Z_o), reactance of C_{oc} , and load resistance (R_L) are connected in series. Hence, the amplified output voltage ($A_v v_i$) will be divided among them. Therefore, the effect of this output coupling capacitor on the frequency response is the same as that of the input coupling capacitor: At low frequencies, there is a significant voltage drop across the capacitive reactance, so less signal is delivered to the load, that results in low voltage gain. The lower cut-off frequency due to C_{oc} is found in the same way as we calculated for C_{ic} . It is the frequency at which the capacitive reactance $X_{C_{oc}}$ will be equal to $Z_o + R_L$. To distinguish these two lower cut-off frequencies, we can represent them as f_{Lic} , and f_{Loc} . The values of these cut-off frequencies are:

$$f_{Lic} = \frac{1}{2\pi(r_s + Z_i)C_{ic}} \text{ Hz} \quad (11-31)$$

$$f_{Loc} = \frac{1}{2\pi(Z_o + R_L)C_{oc}} \text{ Hz} \quad (11-32)$$

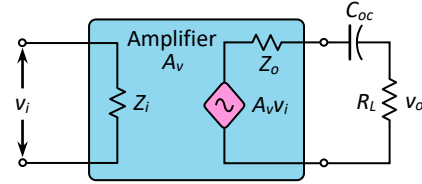


Fig.11-15: Two-port model of an amplifier

Example 11-5

The gain of the amplifier of Fig.11-16 is, $A_v = 256$. Calculate the mid-band overall gain (v_o/v_s) of the amplifier, cut-off frequencies due to the input and output coupling capacitors.

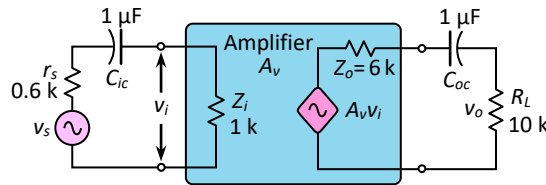


Fig.11-16: Two-port model of an amplifier for Example 11-5

Solution:

For the mid-band frequencies, the coupling capacitors can be considered as short circuits. Therefore,

$$\frac{v_i}{v_s} = \frac{Z_i}{r_s + Z_i} = \frac{1000 \Omega}{(600 + 1000) \Omega} = 0.625$$

and
$$\frac{v_o}{A_v v_i} = \frac{R_L}{Z_o + R_L} = \frac{10 \text{ k}}{6 \text{ k} + 10 \text{ k}} = 0.625$$

or,
$$\frac{v_o}{v_i} = 0.625 \times A_v = 0.625 \times 256 = 160$$

∴ Overall gain,
$$\frac{v_o}{v_s} = \frac{v_i}{v_s} \times \frac{v_o}{v_i} = 0.625 \times 160 = 100 \text{ [Ans.]}$$

or,
$$\frac{v_o}{v_s} (\text{dB}) = 20 \log_{10}(100) \text{ dB} = 40 \text{ dB [Ans.]}$$

Lower cut-off frequency for C_{ic}

Using Equ.(11-31),
$$f_{Lic} = \frac{1}{2\pi(r_s + Z_i)C_{ic}} = \frac{1}{2\pi(600 + 1000)1 \times 10^{-6}} \text{ Hz}$$

∴
$$f_{Lic} = 99.5 \text{ Hz [Ans.]}$$

Lower cut-off frequency for C_{oc}

Using Equ.(11-31),
$$f_{Loc} = \frac{1}{2\pi(Z_o + R_L)C_{oc}} = \frac{1}{2\pi(6000 + 10000)1 \times 10^{-6}} \text{ Hz}$$

∴
$$f_{Loc} = 9.95 \text{ Hz [Ans.]}$$

Comments: The cut-off frequency due to C_{ic} is one decade higher than the cut-off frequency due to C_{oc} though their capacitances are same. This is due to different values of Z_i and Z_o .

If the lower cut-off frequency were determined by C_{Loc} alone, that is, f_{Loc} is much greater than f_{Lic} then the frequency response near the cut-off frequency would have the same shape as that of Fig.11-13. However, normally all capacitors of an amplifier work simultaneously. If both the input and output coupling capacitors are present, the frequency response will be significantly different,

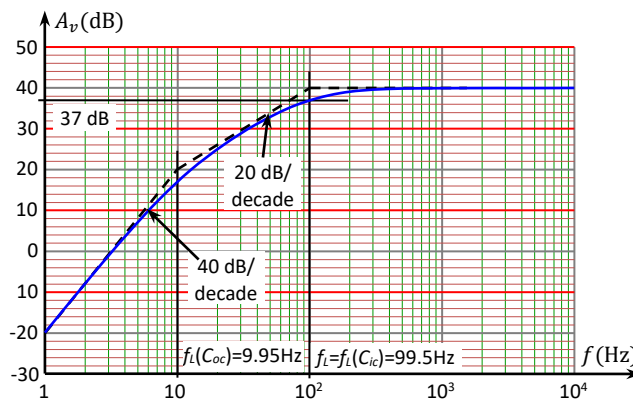


Fig.11-17: Frequency response with two lower cut-off frequencies

because both capacitors reduce the gain simultaneously. The frequency response of the amplifier of Example 10-15 has been redrawn in Fig.11-17. If the two cut-off frequencies are far apart (at

least one decade), the overall lower cut-off frequency, f_L , will be the larger of f_{Lic} and f_{Loc} . Fig.11-17 shows the response of an amplifier that has $f_{Lic} = 99.5$ Hz and $f_{Loc} = 9.95$ Hz with the mid-band gain of 40 dB. From this response, we find that the overall cut-off frequency is $f_L = 99.5$ Hz, because at this point the gain is $40 \text{ dB} - 3 \text{ dB} = 37 \text{ dB}$. At $f_{Loc} = 9.95$ Hz, the overall gain is much less than 37 dB, because at this point both C_{ic} and C_{oc} reduce the gain.

Note that in Fig.11-17 the gain asymptote has two break points one at $f_{Loc} = 9.95$ Hz and the other at $f_{Lic} = 99.5$ Hz. It has a slope of 40 dB/decade (or 12 dB/octave) up to the smaller cut-off frequency $f_{Loc} = 9.95$ Hz, and 20 dB/decade (or 6 dB/octave) up to the larger cut-off frequency $f_{Lic} = 99.5$ Hz.

If the frequencies f_{Lic} and f_{Loc} are closer than one decade, the overall lower cut-off frequency will be somewhat higher than the larger of the two. If $f_{Lic} = f_{Loc}$, the overall lower cut-off frequency will be 1.55 times the value of either one.

Let us draw the frequency response of the amplifier of Example 11-5 considering $C_{oc} = 0.1 \mu\text{F}$. For these case the cut-off frequencies will be: $f_{Lic} = f_{Loc} = 99.5$ Hz. The frequency response is shown in Fig.11-18(a) and the phase shift response is shown in Fig.11-18(b).

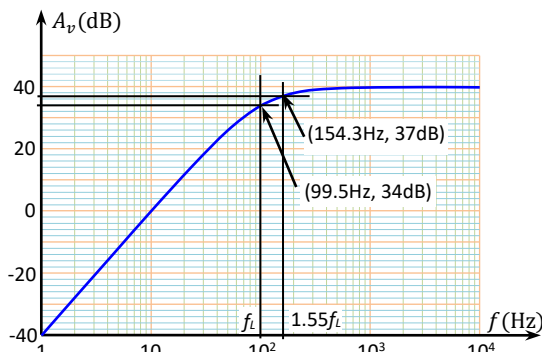


Fig.11-18 (a): Frequency response with $f_{Lic} = f_{Loc} = 99.5$ Hz

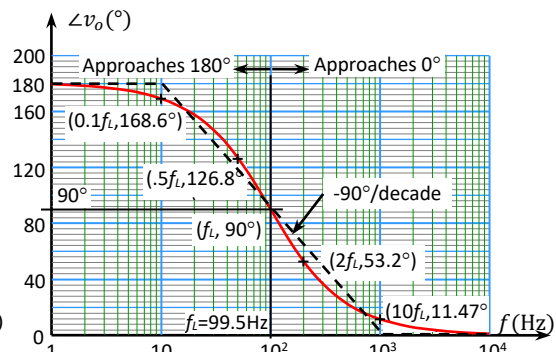


Fig.11-18 (b): Phase response with $f_{Lic} = f_{Loc} = 99.5$ Hz

From Fig.11-18(a), we find that the asymptote breaks downward at $f_{Lic} = f_{Loc} = 99.5$ Hz, and has a slope of 40 dB/decade (or 12 dB/octave). The actual response is 6 dB below the maximum gain (40 dB) at this frequency. But we know that at the lower cut-off frequency the gain will be 3 dB below the maximum gain, which is 37 dB. From the graph, we find, at 37 dB gain the frequency is $f_L = 154.3 \text{ Hz} = 1.55 \times 99.5 \text{ Hz}$. That for this case the cut-off frequency is $1.55 \times f_{Lic}$ or $1.55 \times f_{Loc}$.

As shown in Fig.11-18(b), the phase shift of the output (v_o) with respect to v_s at $f_{Lic} = f_{Loc} = 99.5$ Hz is 90° , and approaches 180° as frequency approaches 0. Remember that this type of response will be found if the RC networks are isolated from each other. Here the amplifier itself isolates the networks.

Effect of Three External Capacitors:

The frequency response due to the bypass capacitor (C_E) will be similar as calculated in Section 11.9.1 [Equ.(11-51)]. If three capacitors work simultaneously, and their cut-off frequencies are far apart, the asymptotic response will have three break points. The overall cut-off frequency will be the largest one. If the largest cut-off frequency is far apart from other two, then also the overall

cut-off frequency will be the largest one. If two cut-off frequencies are equal and much larger from the third one, then the overall cut-off frequency will be 1.55 times the either one of equal cut-off frequencies. However, if the three cut-off frequencies are equal, say $f_{Lic} = f_{Loc} = f_{LE} = f_L$, the overall cut-off frequency will be approximately double of f_L . At $f = f_L$, the gain of the actual response will be 9 dB below the maximum gain.

11.7 Shunt Capacitance and High Frequency Response

The capacitance of amplifier that provides a path for the AC signal to the ground is called **shunt capacitance**. The shunt capacitances formed at the input and output cause reduction in gain at high frequencies. In Chapter 6, we have discussed the inter-electrode capacitances of BJT which are mainly due to device structural capacitances. Shunt capacitances are also created by wiring, terminal electrodes, soldering joint, and any other circuit structure where conducting materials come close together. This type of capacitance is called **stray capacitance**.

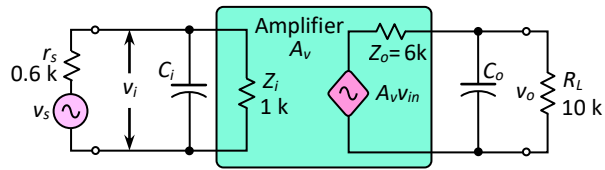


Fig.11-19: Two port model of an amplifier with equivalent input and output capacitances

To study the performance of an amplifier let us represent all input capacitance by C_i and all output capacitance by C_o . Then the simplified diagram of an amplifier will be like the circuit of Fig.11-19.

Effect of Input Shunt Capacitance:

At first we will explain the high frequency response of the amplifier due to the input capacitance, C_i . As shown in Fig.11-20, v_o is open circuit output voltage and $v_o/v_i = A_v$. In the mid-band frequency range, reactance of the shunt capacitances are high enough to be neglected and the reactance of the coupling capacitors are low enough to be neglected. At mid-band frequency range, from Fig.11-20, we can write,

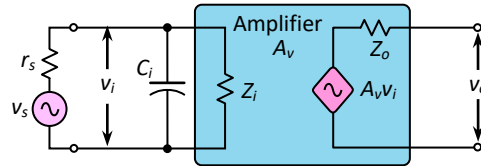


Fig.11-20: Amplifier with input shunt (parasitic) capacitance

$$\frac{v_i}{v_s} = \frac{Z_i}{r_s + Z_i}$$

$$\text{or, } \frac{v_o}{v_s} = \frac{v_i}{v_s} \times \frac{v_o}{v_i} = \frac{Z_i A_v}{r_s + Z_i} = A_{v(\text{mid})} \quad (11-33)$$

For higher frequencies the reactance of C_i will decrease and will be comparable to Z_i . At those frequencies, we have to consider the effect of C_i . So considering the reactance of C_i ($-jX_{Ci}$), we can write

$$\frac{v_i}{v_s} = \frac{-jX_{Ci} \parallel Z_i}{r_s - jX_{Ci} \parallel Z_i} = \frac{(-j/\omega C_i) \parallel Z_i}{r_s + (-j/\omega C_i) \parallel Z_i} \quad [\because X_{Ci} = -j/\omega C_i]$$

$$\text{or, } \frac{v_i}{v_s} = \frac{(-j/\omega C_i)Z_i/[Z_i + (-j/\omega C_i)]}{r_s + (-j/\omega C_i)Z_i/[Z_i + (-j/\omega C_i)]}$$

Multiplying the numerator and denominator of above equation by $[Z_i + (-j/\omega C_i)]$ we get,

$$\frac{v_i}{v_s} = \frac{(-j/\omega C_i)Z_i}{r_s[Z_i + (-j/\omega C_i)] + (-j/\omega C_i)Z_i}$$

Dividing the numerator and denominator of above equation by $(-j/\omega C_i)$ we get,

$$\text{or, } \frac{v_i}{v_s} = \frac{Z_i}{r_s(j\omega C_i Z_i + 1) + Z_i} = \frac{Z_i}{r_s j\omega C_i Z_i + r_s + Z_i}$$

$$\therefore \frac{v_i}{v_s} = \frac{Z_i}{(r_s + Z_i) + j r_s Z_i \omega C_i} \quad (11-34)$$

Magnitude of this equation is,

$$\begin{aligned} \left| \frac{v_i}{v_s} \right| &= \frac{Z_i}{\sqrt{(r_s + Z_i)^2 + (\omega r_s Z_i C_i)^2}} \\ \text{or, } \left| \frac{v_i}{v_s} \right| &= \frac{Z_i/(r_s + Z_i)}{\sqrt{1^2 + [\omega r_s C_i \times Z_i/(r_s + Z_i)]^2}} \\ \text{or, } \left| \frac{v_i}{v_s} \right| &= \frac{Z_i/(r_s + Z_i)}{\sqrt{1 + [\omega C_i \times (r_s \| Z_i)]^2}} \end{aligned} \quad (11-35)$$

Now, the overall gain can be calculated as,

$$\left| \frac{v_o}{v_s} \right| = \frac{v_i}{v_s} \times \frac{v_o}{v_i} = \frac{Z_i/(r_s + Z_i) \times A_v}{\sqrt{1 + [\omega C_i \times (r_s \| Z_i)]^2}} \quad (11-36)$$

$$\text{Using Equ.(11-33), } \left| \frac{v_o}{v_s} \right| = \frac{A_{v(\text{mid})}}{\sqrt{1 + [\omega C_i (r_s \| Z_i)]^2}} \quad (11-37)$$

From this equation, we find that $|v_o/v_s|$ decreases with the increase in frequency. The higher cut-off frequency, ω_{Hi} (i indicates the input capacitance), is the frequency at which the voltage gain falls to $\frac{A_{v(\text{mid})}}{\sqrt{2}}$. Putting this value into Equ.(11-37), we get

$$\begin{aligned} \frac{A_{v(\text{mid})}}{\sqrt{1 + [\omega_{Hi} C_i (r_s \| Z_i)]^2}} &= \frac{A_{v(\text{mid})}}{\sqrt{2}} \\ \text{or, } \frac{1}{\sqrt{1 + [\omega_{Hi} C_i (r_s \| Z_i)]^2}} &= \frac{1}{\sqrt{2}} \\ \text{or, } \sqrt{1 + [\omega_{Hi} C_i (r_s \| Z_i)]^2} &= \sqrt{2} \\ \text{or, } \omega_{Hi} C_i (r_s \| Z_i) &= 1 \\ \therefore \boxed{\omega_{Hi} = \frac{1}{C_i (r_s \| Z_i)} \text{ rad/s}} & \quad (11-38) \end{aligned}$$

and

$$f_{Hi} = \frac{1}{2\pi C_i (r_s \parallel Z_i)} \text{ Hz} \quad (11-39)$$

Gain in terms of f_H :

Using Equ.(11-38) in Equ.(11-37), we get

$$\left| \frac{v_o}{v_s} \right| = |A_v| = \frac{A_{v(\text{mid})}}{\sqrt{1 + (\omega/\omega_{Hi})^2}}$$

 $\therefore$

$$|A_v| = \frac{A_{v(\text{mid})}}{\sqrt{1 + (\omega/\omega_{Hi})^2}} \quad (11-40)$$

and

$$|A_v| = \frac{A_{v(\text{mid})}}{\sqrt{1 + (f/f_{Hi})^2}} \quad (11-41)$$

The shunt capacitance will produce a lagging phase shift with respect to v_s . If we consider $\angle v_s = 0^\circ$, then the angle of v_o (and hence A_v) will be,

$$\angle A_v = -\tan^{-1}(f/f_{Hi}) \quad (11-42)$$

Example 11-6

The open circuit voltage gain of the amplifier of Fig.11-21 is $v_i/v_o = 160$. Calculate the overall mid-band voltage gain and the higher cut-off frequency.

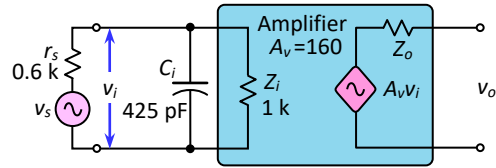


Fig.11-21: Amplifier with input shunt capacitance for Example11-6

Solution:

Using Equ.(11-33),

$$A_{v(\text{mid})} = \frac{1000 \times 160 \Omega}{(600 + 1000) \Omega} = 100 \text{ [Ans.]}$$

or,

$$A_{v(\text{mid})}(\text{dB}) = 20 \log_{10} 100 = 40 \text{ dB [Ans.]}$$

Here,

$$r_s \parallel Z_i = \frac{r_s Z_i}{r_s + Z_i} = \frac{600 \times 1000}{600 + 1000} \Omega = 375 \Omega$$

Using Equ.(11-39),

$$f_{Hi} = \frac{1}{2\pi(425 \times 10^{-12}) \times 375} \text{ Hz} \approx 1 \text{ MHz [Ans.]}$$

Comments: As the output terminal is open, Z_o has no effect on the gain. But, if a load is connected at the output, the gain will decrease due to the output impedance, Z_o .

The high frequency response of the amplifier of Example 11-6 are plotted in Fig.11-22. Fig.11-22(a) shows the variation of voltage gain in dB with frequency in a semi-log graph, and Fig.11-22(b) shows the variation of phase angel of v_i with respect to v_s with the variation in frequency. The

mid-band voltage gain ($A_{v(\text{mid})}$) and the higher cut-off frequency (f_{Hi}) of the amplifier is 100 (or 40 dB) and 1 MHz, respectively. As shown in Fig.11-22(a), at the cut-off frequency, the gain is 37 dB, that is 3 dB below the maximum gain. After this cut-off frequency, the gain decreases at a rate of 20 dB/decade or 6 dB/octave. **One octave above the cut-off frequency the gain is 89.5% of mid-band gain, and one octave below it is 44.7%. Again, one decade above f_L the gain is 99.5% of mid-band gain, and one decade below it is 9.95%.** That is, the high frequency response is exactly same as the low frequency response. Only the difference is, here gain decreases with frequency. Fig.11-22(b) shows that the phase shift is 45° at the cut-off frequency. One octave above the cut-off frequency the phase shift is 26.6° , and one octave below it is 63.4° . Again, one decade above the cut-off frequency the phase shift is 5.7° , and one decade below it is 84.3° . As shown in Fig.11-22(a), the asymptotic response breaks at the point $f = f_{Hi}$. For this reason f_{Hi} is also called break point frequency. The slope of the asymptote is 6 dB/octave or 20 dB/decade.

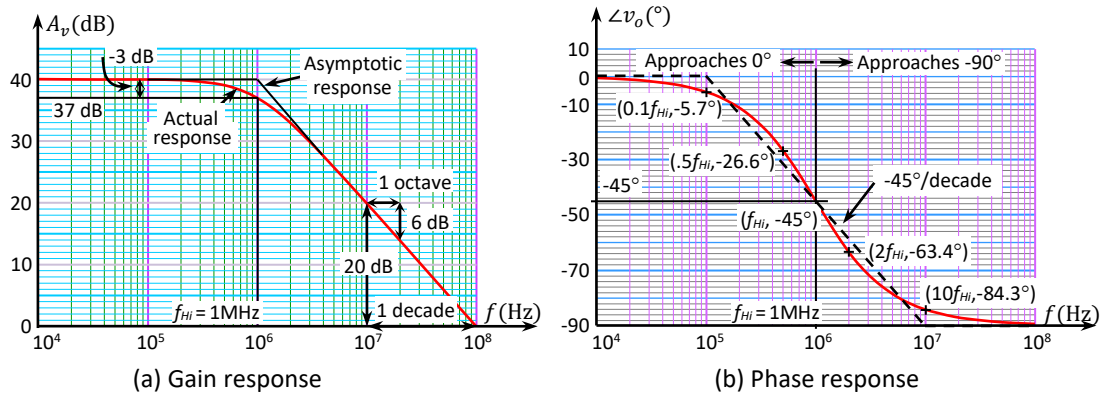


Fig.11-22: High frequency response with input shunt capacitance

Effect of Output Shunt Capacitance:

If we consider that the amplifier has no capacitance in the input side and $r_s = 0$, then the effect of the output capacitance will be found. The circuit is shown in Fig.11-23. Here also, in the mid-band frequency range, C_o will be open, so

$$v_o = \frac{(A_v v_i) \times R_L}{Z_o + R_L}$$

or

$$\frac{v_o}{v_i} = \frac{R_L A_v}{Z_o + R_L} \quad (11-43)$$

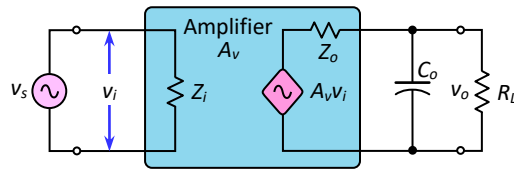


Fig.11-23: Amplifier with output shunt capacitance

And

$$v_i = v_s \quad \text{or,} \quad \frac{v_i}{v_s} = 1 \quad (11-44)$$

Therefore, the overall mid-band voltage gain will be,

$$A_{v(\text{mid})} = \frac{v_o}{v_s} = \frac{v_i}{v_s} \times \frac{v_o}{v_i} = 1 \times \frac{Z_i A_v}{r_s + Z_i} \quad (11-45)$$

Here also, using the same analysis of the input shunt capacitance, we can show that, the higher cut-off frequency (f_{Ho} , C_o means output capacitance) is,

$$\omega_{Ho} = \frac{1}{C_o(Z_o \parallel R_L)} \text{ rad/s} \quad (11-46)$$

or,

$$f_{Ho} = \frac{1}{2\pi C_o(Z_o \parallel R_L)} \text{ Hz} \quad (11-47)$$

In terms of the cut-off frequency, the gain will be

$$|A_v| = \frac{A_{v(\text{mid})}}{\sqrt{1 + (f/f_{Ho})^2}}$$

This output shunt capacitance will also produce a lagging phase shift from v_s . If we consider $\angle v_s = 0^\circ$, then the angle of v_o (and hence A_v) will be,

$$\angle A_v = -\tan^{-1}(f/f_{Ho})$$

In any amplifier the input and output shunt capacitances work simultaneously. The high frequency response due to both the capacitances will be same as the two coupling capacitors discussed earlier in section 11.6. If the cut-off frequencies are far apart (at least one decade), the response will have two break points. The overall higher cut-off frequency will be the smaller one. If the two cut-off frequencies are equal, the overall higher cut-off frequency will be 1.55 times the either one. The situation is explained further in Example 11-7.

Example 11-7

For the amplifier of Fig.11-24, calculate the overall mid-band gain, cut-off frequencies due the input and output shunt capacitances and plot the frequency response (Bode plot).

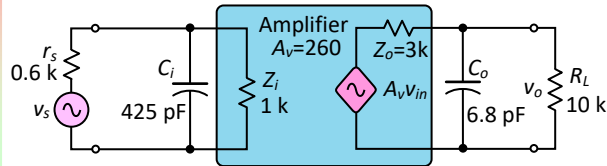


Fig.11-24: Amplifier with input and output shunt capacitance for Example 11-7

Solution:

For mid-band frequencies, both the input and output capacitances will be open. So, the overall mid-band gain of the amplifier will be,

$$A_{v(\text{mid})} = A_v \times \frac{Z_i}{r_s + Z_i} \times \frac{R_L}{Z_o + R_L}$$

$$\text{or,} \quad A_{v(\text{mid})} = 260 \times \frac{1000 \Omega}{(600 + 1000) \Omega} \times \frac{10 \text{ k}}{(3 + 10) \text{ k}}$$

$$\therefore A_{v(\text{mid})} = 125 \approx 42 \text{ dB [Ans.]}$$

Here,

$$r_s \parallel Z_i = \frac{r_s Z_i}{r_s + Z_i} = \frac{600 \times 1000}{600 + 1000} \Omega = 375 \Omega$$

Higher cut-off frequency due to C_i ,

Using Equ.(11-39)

$$f_{Hi} = \frac{1}{2\pi(425 \times 10^{-12}) \times 375} \text{ Hz} \approx 1 \text{ MHz [Ans.]}$$

Higher cut-off frequency due to C_o ,

Using Equ.(11-47),

$$f_{Ho} = \frac{1}{2\pi(6.8 \times 10^{-12})(3 \text{ k} \parallel 10 \text{ k})} \approx 10 \text{ MHz [Ans.]}$$

The high frequency response of this amplifier can be drawn using the following expression,

$$|A_v| = A_{v(\text{mid})} \times \frac{1}{\sqrt{1 + (f/f_{Hi})^2}} \times \frac{1}{\sqrt{1 + (f/f_{Ho})^2}}$$

The graph of this equation is shown in Fig.11-25.

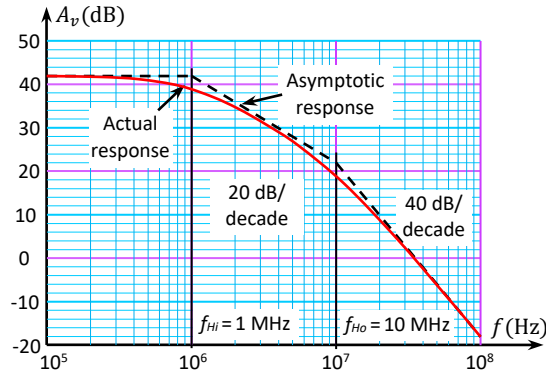


Fig.11-25: High frequency response with C_i and C_o

Comments: The frequency response shows that it has two corner frequencies due to the input and output parasitic capacitances.

11.8 Bandwidth and Gain-Bandwidth Product

The range of frequencies between the lower and higher cut-off frequencies is called **bandwidth** (BW). This is the range of frequencies that the amplifier can properly amplify. Mathematically,

$$\text{BW} = f_H - f_L \quad (11-48)$$

In many amplifiers, like audio amplifiers, the lower cut-off frequency is very small compared to the higher cut-off frequency. In that case, the higher cut-off frequency will be the bandwidth of the amplifier. It is shown in Fig.11-26.

The product of gain (generally voltage gain) and the bandwidth of an amplifier is called the **gain-bandwidth product**. It is a figure of merit of an amplifier, because it provides the information about the gain of the amplifier and its operating frequency range. For a particular amplifier the gain-bandwidth product is constant. So, if we increase the gain of the amplifier the bandwidth will decrease and vice versa.

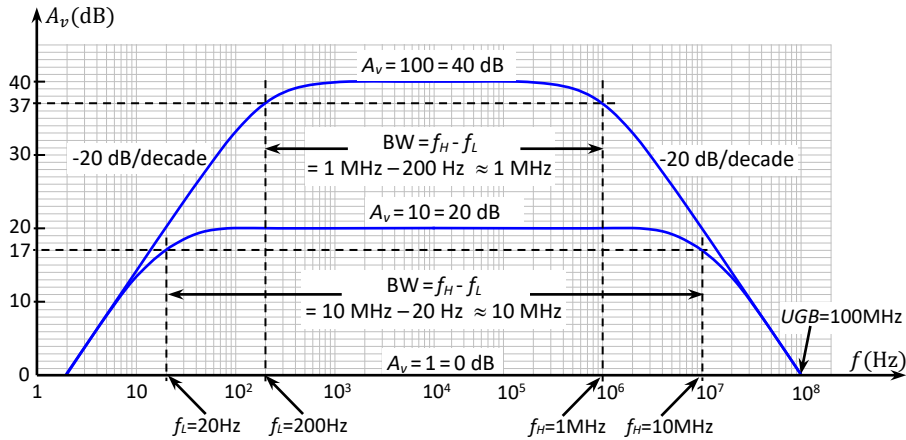


Fig.11-26: Typical bandwidth of an amplifier

Fig.11- 26 shows the frequency response of an amplifier with a voltage gain $A_v = 100$ or 40 dB, lower cut-off frequency $f_L = 200$ Hz, and higher cut-off frequency $f_H = 1$ MHz. Therefore, the bandwidth of the amplifier will be,

$$BW = f_H - f_L = 1 \text{ MHz} - 200 \text{ Hz} \approx 1 \text{ MHz}$$

and the gain-bandwidth product will be,

$$GBP = A_v \times BW = 1 \text{ MHz} \times 100 = 100 \text{ MHz}$$

On the other hand, if the gain of the amplifier is decreased to $A_v = 10$ or 20 dB, the bandwidth will be approximately 10 MHz as shown in the same figure. Therefore, the new GBP will be $10 \text{ MHz} \times 10 = 100 \text{ MHz}$, same as before. Moreover, if the gain of the amplifier is set to *unity*, the gain-bandwidth product will be 100 MHz which is called **unity gain bandwidth** (UGB). Note the response of the amplifier: it has only one break point on the low frequency and high frequency side. For amplifiers having only one break point in both sides, the gain-bandwidth product is always constant, as found here.

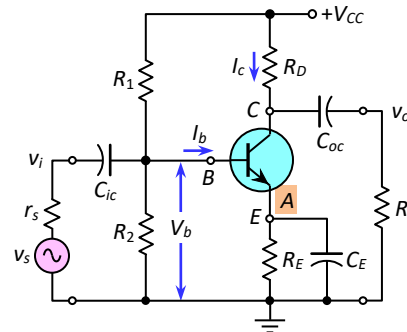


Fig.11-27: CE amplifier with voltage divider

11.9 Frequency Response of Common Emitter Amplifier

The circuit diagram of a common emitter amplifier is given in Fig.11-27. The AC equivalent circuit can be drawn using either r_e -model or hybrid model of transistor as described in Chapter 6. Here we will use the hybrid modeling.

Low Frequency Response

The analysis of this amplifier has been covered in Section 6.12.1 of Chapter 6. For better understating the mid-frequency equivalent circuit is redrawn in Fig.11-28. In that analysis we have calculated the voltage gain, input impedance, output impedance, and current gain. The values are given below.

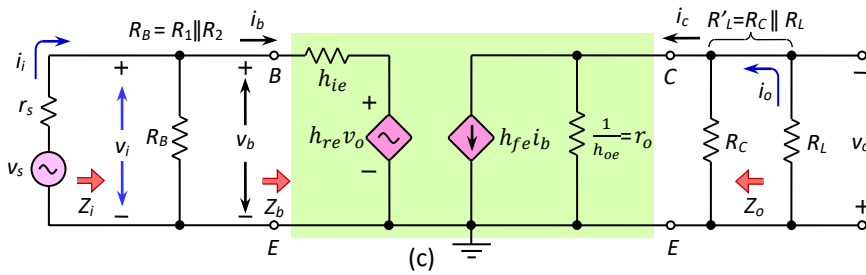


Fig.11-28: Mid-frequency equivalent circuit of CE amplifier

From Equ.(6-56),

$$A_v \approx -\frac{h_{fe} R'_L}{h_{ie}} = -\frac{h_{fe}}{h_{ie}} \left(\frac{R_C R_L}{R_C + R_L} \right)$$

From Equ.(6-59),

$$Z_i = R_B || Z_b = R_B || h_{ie}$$

From Equ.(6-62),

$$Z_o = r_o || R_C \approx R_C$$

As discussed in Section 11.6.1, the lower cut-off frequency of this amplifier due to the input coupling capacitor will be,

From Equ.(11-22),

$$f_{Lic} = \frac{1}{2\pi(r_s + Z_i)C_{ic}}$$

$$\therefore f_{Lic} = \frac{1}{2\pi(r_s + R_B || h_{ie})C_{ic}} \text{ Hz} \quad (11-49)$$

The lower cut-off frequency of due to the output coupling capacitor will be,

From Equ.(11-32),

$$f_{Loc} = \frac{1}{2\pi(Z_o + R_L)C_{oc}}$$

$$\therefore f_{Loc} = \frac{1}{2\pi(r_o || R_C + R_L)C_{oc}} \text{ Hz} \quad (11-50)$$

The amplifier that uses bypass capacitor will have another lower cut-off frequency due to this bypass capacitor. To calculate this cut-off frequency, we have to calculate the resistance seen by C_E capacitor at point A. $r_s || R_B$ will come in series with the base-to-emitter resistance (h_{ie}). That is

total resistance in the base side will be $r_s \parallel R_B + h_{ie}$. When this resistance is seen from the emitter side it will be divided by h_{fe} and come in parallel with R_E . Therefore, the of total resistance seen at point A will be,

$$R_{\text{equi}} = \left(\frac{r_s \parallel R_B + h_{ie}}{h_{fe}} \right) \parallel R_E \quad (11-51)$$

Where, $R_B = R_1 \parallel R_2$. The cut-off frequency will occur when the reactance of C_E will be equal to R_{equi} . That is,

$$X_{C_E} = R_{\text{equi}} \text{ [at } f = f_{LE}]$$

$$\text{or,} \quad \frac{1}{2\pi f_{LE} C_E} = R_{\text{equi}}$$

$$\text{or,} \quad f_{LE} = \frac{1}{2\pi R_{\text{equi}} C_E}$$

Using Equ.(11-51) we get,

$$f_{LE} = \frac{1}{2\pi [(r_s \parallel R_B + h_{ie}) / h_{fe} \parallel R_E] C_E} \quad (11-52)$$

Among the three cut-off frequencies, the overall lower cut-off frequency (f_L) of the amplifier has to be determined as described in Section 11.6.2. If the frequencies are far apart, or one frequency is much larger than the other two, f_L will be the larger one. If two of them are equal and they are larger than the third one, f_L will be 1.55 times either one of the equal frequencies. If all are equal, then f_L will be double on any value. However, generally in CE amplifier the lower cut-off frequency is set by the bypass capacitor C_E as this capacitor needs the largest capacitance.

Example 11-8

Determine the lower cut-off frequencies imposed by three coupling and bypass capacitors of the CE amplifier given in Fig.11-29.

Solution:

To determine r_e we need the value of I_E . To calculate I_E let us check the condition $\beta R_E \gg 10R_2$. $\beta R_E = 50 \times 2.4 \text{ k} = 120 \text{ k}$, and $10R_2 = 10 \times 27 \text{ k} = 270 \text{ k}$. The condition is not satisfied. So, we have to use exact analysis.

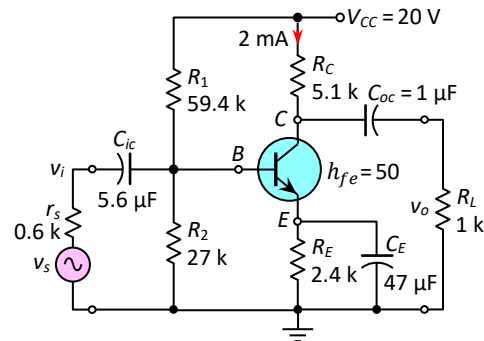


Fig.11-29: CE amplifier for Example 11.8

$$\text{Using Equ.(6-55),} \quad R_{\text{Th}} = \frac{R_1 R_2}{R_1 + R_2} = \frac{59.4 \text{ k} \times 27 \text{ k}}{59.4 \text{ k} + 27 \text{ k}} \approx 18.56 \text{ k} = R_B$$

$$\text{Using Equ.(6-54),} \quad E_{\text{Th}} = \frac{V_{CC} R_2}{R_1 + R_2} = \frac{20 \text{ V} \times 27 \text{ k}}{59.4 \text{ k} + 27 \text{ k}} \approx 6.25 \text{ V}$$

Using Equ.(6-57),
$$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (h_{fe} + 1)R_E} = \frac{6.25 \text{ V} - 0.7 \text{ V}}{18.56 \text{ k} + (50 + 1)2.4 \text{ k}} \approx 39.373 \mu\text{A}$$

Collector current
$$I_C = h_{fe}I_B = 39.373 \mu\text{A} \times 50 = 1.97 \text{ mA} \approx I_E$$

$$\therefore r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{1.97 \text{ mA}} = 13.2 \Omega$$

$$\therefore h_{ie} = h_{fe}r_e = 50 \times 13.2 \Omega = 0.66 \text{ k}$$

Using Equ.(11-49),
$$f_{Lic} = \frac{1}{2\pi(0.6 \text{ k} + 18.56 \text{ k} \parallel 0.66 \text{ k}) \times 5.6 \mu\text{F}} \approx 23 \text{ Hz}$$

Using Equ.(11-50),
$$f_{Loc} = \frac{1}{2\pi(200 \text{ k} \parallel 5.1 \text{ k} + 1 \text{ k}) \times 1 \mu\text{F}} \approx 32 \text{ Hz}$$

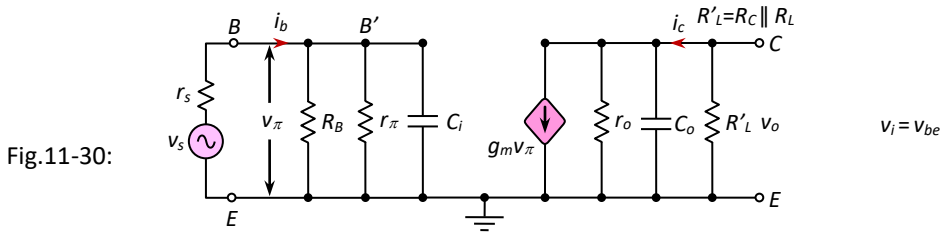
Using Equ.(11-52),
$$f_{LE} = \frac{1}{2\pi[(0.6 \text{ k} \parallel 18.56 \text{ k} + 0.66 \text{ k})50 \parallel 2.4 \text{ k}] \times 47 \mu\text{F}} \approx 137 \text{ Hz}$$

As, 137 Hz is much (≈ 1 decade) larger than the other two, the effective lower cut-off frequency of the amplifier will be $f_L = 137 \text{ Hz}$.

Comments: In Chapter 10, we have found that for CE amplifier the bypass capacitor (C_E) determines the lower cut-off frequency. The same result is found here.

11.9.1 High Frequency Response

By using the hybrid- π model of transistor, we can determine the high frequency response of this amplifier. The hybrid- π models of BJT have been discussed in Chapter 6. Using that hybrid- π model we can draw the equivalent circuit of CE amplifier as shown in Fig.11-30.



Here, C_i is the total high frequency capacitance in the input side. It includes the base emitter junction capacitance (C_{π}), Miller capacitance (C_{Mi}) and the wiring capacitance of the circuit (C_{wi}). The voltage gain of the amplifier is $-g_m r_o \parallel R'_L$ and $R'_L = R_C \parallel R_L$. Therefore, the value of the Miller capacitance will be $C_{Mi} = C_{\mu}(1 + g_m r_o \parallel R'_L)$. That is,

$$C_i = C_{\pi} + C_{Mi} + C_{wi} = C_{\pi} + C_{\mu}(1 + g_m r_o \parallel R'_L) + C_{wi} \quad (11-53)$$

Similarly, C_o is the total high frequency capacitance in output side. It includes the collector emitter junction capacitance (C_{ce}), Miller capacitance (C_{Mo}) and the wiring capacitance of the circuit (C_{wo}). Value of the Miller capacitance will be $C_{Mo} = C_{\mu}(1 + g_m r_o \parallel R'_L)/(g_m r_o \parallel R'_L) \approx C_{\mu}$. That is,

$$C_o = C_{ce} + C_{Mo} + C_{wo} = C_{ce} + C_{\mu} + C_{wo} \quad (11-54)$$

All of these input and output capacitances are shown in Fig.11-31, to understand them at a glance.

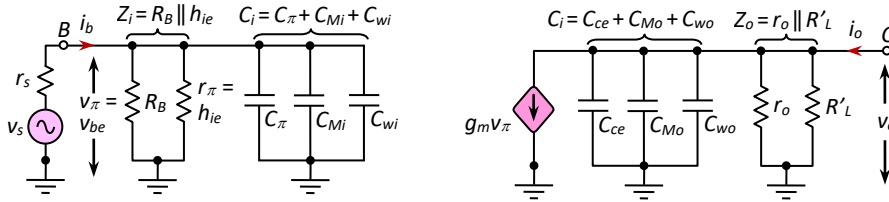


Fig.11-31: All input and output parasitic capacitances of CE amplifier

We have already derived the higher cut-off frequencies for the input and output capacitance in topic 11-7. The higher cut-off frequency due to the input capacitance is given by Equ.(11-39),

$$f_{Hi} = \frac{1}{2\pi C_i (r_s \parallel Z_i)} \text{ Hz}$$

Here, the value of input impedance will be,

$$Z_i = R_B \parallel r_{\pi} \approx R_B \parallel h_{ie}$$

$$\therefore \boxed{f_{Hi} = \frac{1}{2\pi C_i (r_s \parallel R_B \parallel h_{ie})} \text{ Hz}} \quad (11-55)$$

If R_B is very large,

$$\boxed{f_{Hi} = \frac{1}{2\pi C_i (r_s \parallel h_{ie})} \text{ Hz} = \frac{r_s + h_{ie}}{2\pi C_i r_s h_{ie}} \text{ Hz}} \quad (11-56)$$

Similarly the higher cut-off frequency due to the output capacitance is given by Equ.(11-47),

$$f_{Ho} = \frac{1}{2\pi C_o (Z_o \parallel R_L)} \text{ Hz}$$

Here, the value of output impedance will be,

$$Z_o = r_o \parallel R'_L = r_o \parallel R_C \parallel R_L$$

$$\therefore \boxed{f_{Ho} = \frac{1}{2\pi C_o (r_o \parallel R_C \parallel R_L)} \text{ Hz}} \quad (11-57)$$

$$\text{As, } r_o \gg R_C \parallel R_L \quad \boxed{f_{Ho} = \frac{1}{2\pi C_o (R_C \parallel R_L)} \text{ Hz} = \frac{R_C + R_L}{2\pi C_o (R_C R_L)} \text{ Hz}} \quad (11-58)$$

The value of f_{Hi} and f_{Ho} will produce two break points on the asymptote at the high frequency region like the response of Fig.11-25. Here also, the overall higher cut-off frequency (f_H) of the amplifier has to be determined as described in Section 11-17. If the frequencies (break points) are far apart, then f_H will be the smaller one. If they are equal, then f_H will be 1.55 times either one.

The cut-off frequencies f_{Hi} and f_{Ho} are produced due to the parasitic capacitances of the device and circuit. However, they are not the only parameters that limit the high frequency response of the amplifiers. The value of h_{fe} (or β) of a transistor decreases with frequency in the high

frequency region. Therefore, in addition to f_{Hi} and f_{Ho} , we must consider the high frequency limitation imposed by the decrease of h_{fe} with frequency. In fact, h_{fe} will produce another higher cut-off frequency (discussed below). Generally, this cut-off frequency would be very larger than f_{Hi} and f_{Ho} . If this cut-off frequency is comparable to f_{Hi} and f_{Ho} , then the overall higher cut-off frequency of the amplifier has to be determined considering all of the three cut-off frequencies.

But generally for CE amplifier f_{Hi} becomes the higher cut-off frequency of the amplifier, because the value of C_i is larger than C_o due to the Miller effect.

As the value of lower cut-off frequency of an amplifier is very small compared to the higher cut-off frequency, we can consider the higher cut-off frequency as the bandwidth of the amplifier. Therefore, the bandwidth of the CE amplifier will be,

$$\therefore BW = f_{Hi} = \frac{1}{2\pi C_i (r_s \parallel R_B \parallel h_{ie})} \text{ Hz} \approx \frac{r_s + h_{ie}}{2\pi C_i r_s h_{ie}} \text{ Hz} \quad (11-59)$$

and the gain-bandwidth product will be,

$$\therefore GBP = A_v \times BW = \frac{h_{fe}}{h_{ie}} \left(\frac{R_C R_L}{R_C + R_L} \right) \times \frac{r_s + h_{ie}}{2\pi C_i r_s h_{ie}} \text{ Hz} \quad (11-60)$$

Example 11-9

Determine the upper cut-off frequency of the amplifier of Example 11-8. Assume that the transistor used in the amplifier has $C_{be} = 10$ pF, $C_{bc} = 5$ pF, and $C_{ce} = 2$ pF. $r_o = 200$ k. Neglect wiring capacitances.

Solution:

Here, $C_{be} = C_\pi = 10$ pF, $C_{bc} = C_\mu = 5$ pF and in Example 10-, we have found $r_e = 13.2 \Omega$

We know that,

$$g_m \approx \frac{1}{r_e} = \frac{1}{13.2 \Omega} \approx 75.76 \text{ mS}$$

Voltage gain of the amplifier will be,

$$A_v = g_m r_o \parallel R_C \parallel R_L = 75.76 \text{ mS} \times (200 \text{ k} \parallel 5.1 \text{ k} \parallel 1 \text{ k}) \approx 63$$

$$\text{Using Equ.(11-53), } C_i = 10 \text{ pF} + 5 \text{ pF}(1 + 63) + 0 = 333 \text{ pF}$$

$$\text{Using Equ.(11-54), } C_o = C_{ce} + C_{Mo} + C_{wo} = 2 \text{ pF} + 5 \text{ pF} + 0 = 7 \text{ pF}$$

$$\text{Using Equ.(11-55), } f_{Hi} = \frac{1}{2\pi \times 333 \text{ pF}(0.6 \text{ k} \parallel 18.56 \text{ k} \parallel 0.66 \text{ k})} \approx 1.55 \text{ MHz}$$

$$\text{Using Equ.(11-58), } f_{Ho} = \frac{1}{2\pi \times 7 \text{ pF}(200 \text{ k} \parallel 5.1 \text{ k} \parallel 1 \text{ k})} \text{ Hz} \approx 27 \text{ MHz}$$

Between these two frequencies, the effective higher cut-off frequency will be the smaller one, that is $f_H = 1.5 \text{ MHz}$.

Comments: In CE amplifier generally, the input parasitic capacitance determine the higher cut-off frequency, due to the Miller's effect.

11.9.2 Beta Cut-off Frequency

Beta cut-off frequency is a parameter related with the device not with the circuit. It is a measure of how the device can amplify the high frequencies. It is defined as the frequency at which the short-circuit current gain of a CE amplifier or a BJT in CE configuration falls by 3 dB from the mid-band (at 1 kHz) current gain. To determine this value, we have to short the output terminal of the circuit of Fig.11-30 redrawn in Fig.11-32 with the output short.

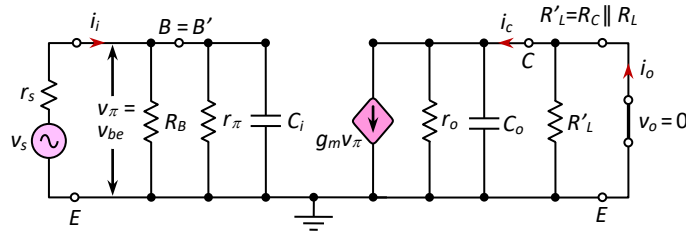


Fig.11-32: High frequency equivalent circuit of CE amplifier

As the output terminal is short all the current ($g_m v_\pi$) will flow through that short terminal. Therefore,

$$i_o = g_m v_\pi = g_m v_{be} \quad (11-61)$$

Value of the input current will be,

$$i_i = \frac{v_\pi}{R_B \parallel r_\pi \parallel X_{Ci}}$$

As beta cut-off frequency is a device property not the circuit properly, we will neglect R_B . So,

$$i_i = \frac{v_\pi}{r_\pi \parallel X_{Ci}}$$

Using $X_{Ci} = -j/\omega C_i$

$$i_i = \frac{v_\pi}{r_\pi(-j/\omega C_i)/[r_\pi + (-j/\omega C_i)]} = \frac{v_\pi[r_\pi + (-j/\omega C_i)]}{r_\pi(-j/\omega C_i)}$$

Dividing the numerator and denominator of the above equation by $(-j/\omega C_i)$ we get,

$$i_i = \frac{v_\pi[r_\pi/(-j/\omega C_i) + 1]}{r_\pi} = \frac{v_\pi[j\omega r_\pi C_i + 1]}{r_\pi} \quad (11-62)$$

Therefore, the short-circuit current gain can be calculated using Equ.(11-61) and Equ.(11-62) as,

$$A_{ish} = \frac{i_o}{i_i} = g_m v_\pi \times \frac{r_\pi}{v_\pi(j\omega r_\pi C_i + 1)} = \frac{g_m r_\pi}{j\omega r_\pi C_i + 1} \quad (11-63)$$

Magnitude of this gain will be,

$$|A_{ish}| = \frac{g_m r_\pi}{\sqrt{(\omega r_\pi C_i)^2 + 1}} \quad (11-64)$$

But, $r_\pi = h_{ie} = \beta r_e$ and $g_m = 1/r_e$. $\therefore g_m r_\pi = (1/r_e)\beta r_e = \beta$

$$\therefore |A_{ish}| = \frac{g_m r_\pi}{\sqrt{(\omega r_\pi C_i)^2 + 1}} = \frac{\beta}{\sqrt{(\omega r_\pi C_i)^2 + 1}} \quad (11-65)$$

At mid-band frequency, the parasitic capacitance will be open, that is

$$\therefore \frac{1}{\omega C_i} = \infty \therefore \omega C_i = 0$$

Therefore, the mid-band current gain will be,

$$|A_{ish(\text{mid})}| = g_m r_\pi = \beta \quad (11-66)$$

At higher cut-off frequency the gain will fall to $|A_{ish(\text{mid})}|/\sqrt{2}$. The frequency at which the short-circuit current gain falls to $A_{ish(\text{mid})}/\sqrt{2}$, is called **beta cut-off frequency** and is denoted by f_β . Therefore,

$$|A_{ish}| = \frac{\beta}{\sqrt{(\omega_\beta r_\pi C_i)^2 + 1}} = \frac{A_{ish(\text{mid})}}{\sqrt{2}} = \frac{\beta}{\sqrt{2}}$$

$$\text{or,} \quad \sqrt{(\omega_\beta r_\pi C_i)^2 + 1} = \sqrt{2}$$

$$\text{or,} \quad (\omega_\beta r_\pi C_i)^2 + 1 = 2$$

$$\text{or,} \quad (\omega_\beta r_\pi C_i)^2 = 1$$

$$\text{or,} \quad \omega_\beta r_\pi C_i = 1$$

$$\therefore \boxed{\omega_\beta = \frac{1}{r_\pi C_i} = \frac{1}{h_{ie} C_i}} \quad (11-67)$$

As the output is shorted the voltage gain ($g_m r_o \| R'_L$) will be zero, because $r_o \| R'_L = 0$. Therefore, Equ.(11-53) can be written as,

$$\therefore C_i = C_\pi + C_\mu(1 + g_m r_o \| R'_L) + C_{wi} = C_\pi + C_\mu + C_{wi}$$

As the wiring capacitance, C_{wi} is very small, neglecting this capacitance we get,

$$C_i = C_\pi + C_\mu \quad (11-68)$$

Putting this value into Equ.(11-67) we get,

$$\therefore \boxed{\omega_\beta = \frac{1}{r_\pi(C_\pi + C_\mu)} \approx \frac{1}{h_{ie}(C_\pi + C_\mu)}} \quad (11-69)$$

$$\text{or, } f_{\beta} = \frac{1}{2\pi r_{\pi}(C_{\pi} + C_{\mu})} \approx \frac{1}{2\pi h_{ie}(C_{\pi} + C_{\mu})} \quad (11-70)$$

As, $h_{ie} = h_{fe}r_e$, and r_e depends on I_E that is on the DC biasing condition, we can say that f_{β} also depends on the biasing condition.

Now, the short-circuit current gain of Equ.(11-65) can be written in terms of f_{β} as,

$$|A_{ish}| = \frac{g_m r_{\pi}}{\sqrt{(\omega r_{\pi} C_i)^2 + 1}} = \frac{|A_{ish(mid)}|}{\sqrt{[\omega/(1/r_{\pi} C_i)]^2 + 1}} \quad \text{Using Equ.(11-66)}$$

$$\text{or, } |A_{ish}| = \frac{|A_{ish(mid)}|}{\sqrt{(\omega/\omega_{\beta})^2 + 1}} \quad \text{Using Equ.(11-67)}$$

$$\text{or, } |A_{ish}| = \frac{|A_{ish(mid)}|}{\sqrt{1 + (f/f_{\beta})^2}} \quad \text{Using Equ.(11-66)}$$

Using Equ.(11-66)

$$|A_{ish}| = \frac{\beta}{\sqrt{1 + (f/f_{\beta})^2}}$$

This equation represents how $|A_{ish}|$ changes with frequency. The variation of $|A_{ish}|$ with f is shown in Fig.11-33.

11.9.3 Unity Current Gain Frequency

In the previous we have discussed about beta cut-off frequency. Here we will discuss about another device parameter called unity current gain frequency. We have already seen that at $f = f_{\beta}$, the short circuit current gain will decrease by 3 dB. If the signal frequency (f) is increased further, at some frequency, $f = f_T$, the short circuit current gain will be 1 (unity). This frequency is called **unity current gain frequency** (f_T). f_T will also be equal to the gain-bandwidth product. Using $|A_{ish}| = 1$ in Equ.(11-65) we get,

$$|A_{ish}| = \frac{g_m r_{\pi}}{\sqrt{(\omega_T r_{\pi} C_i)^2 + 1}} = 1$$

$$\text{or, } \sqrt{(\omega_T r_{\pi} C_i)^2 + 1} = g_m r_{\pi}$$

$$\text{or, } (\omega_T r_{\pi} C_i)^2 = (g_m r_{\pi})^2 - 1$$

$$\text{or, } (\omega_T r_{\pi} C_i)^2 \approx (g_m r_{\pi})^2$$

$$\text{or, } \omega_T r_{\pi} C_i = g_m r_{\pi}$$

$$\therefore \omega_T = \frac{g_m}{C_i} \quad (11-71)$$

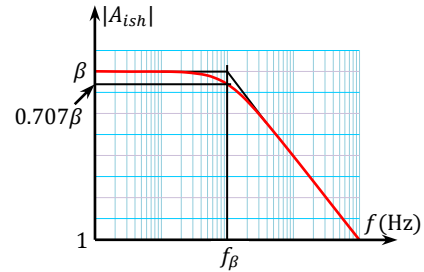


Fig.11-33: Variation of short-circuit current gain with frequency

Here also we can use $C_i = C_\pi + C_\mu$ from Equ.(11-68). Therefore,

$$\omega_T = \frac{g_m}{C_\pi + C_\mu} \quad (11-72)$$

$$\text{or,} \quad f_T = \frac{g_m}{2\pi(C_\pi + C_\mu)} = \frac{1}{2\pi r_e(C_\pi + C_\mu)} \quad [\because g_m = 1/r_e] \quad (11-73)$$

Dividing Equ.(11-73), by Equ.(11-70), we get,

$$\frac{f_T}{f_\beta} = \frac{1}{2\pi r_e(C_\pi + C_\mu)} \times \frac{2\pi h_{ie}(C_\pi + C_\mu)}{1} = \frac{h_{ie}}{r_e} = \beta$$

$$\therefore \quad f_T = \beta f_\beta \quad (11-74)$$

Generally, in datasheets of BJT, the value of f_T is mentioned. Note that f_T is the frequency at which the short-circuit current gain of the transistor becomes unity. Therefore, we can say f_T is gain-bandwidth product in case of device current gain. However, UGB described in Section 11.8 is the gain-bandwidth product in case of circuit voltage gain.

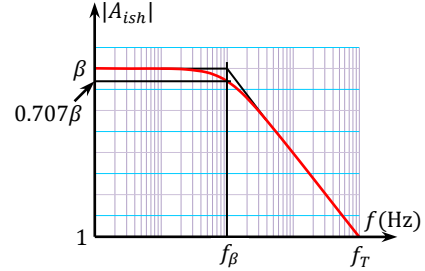


Fig.11-34: Position of unity current gain frequency f_T

Example 11-10

For the transistor of Example 11-8 & 11-9 determine the values of f_β , and f_T .

Solution:

In previous two examples we found $h_{ie} = 0.66 \text{ k}$, $g_m = 75.76 \text{ mS}$, and $C_i = 333 \text{ pF}$.

$$\text{Using Equ.(11-70),} \quad f_\beta \approx \frac{1}{2\pi h_{ie}(C_\pi + C_\mu)}$$

$$\text{or,} \quad f_\beta = \frac{1}{2\pi \times 0.66 \text{ k} \times (10 + 5) \text{ pF}} = 16 \text{ MHz [Ans.]}$$

$$\text{Using Equ.(11-73),} \quad f_T = \frac{g_m}{2\pi(C_\pi + C_\mu)}$$

$$\text{or,} \quad f_T = \frac{75.76 \text{ mS}}{2\pi \times (10 + 5) \text{ pF}} \approx 803.84 \text{ MHz}$$

Comments: $803.84 \text{ MHz} \div 16 \text{ MHz} \approx 50$. This result confirms $f_T = h_{fe} \times f_\beta$.

11.10 Frequency Response of Common Base Amplifier

The circuit diagram of a common base amplifier with voltage divider bias is given in Fig.11-35. In this circuit the inter-electrode capacitances are also shown.

AC analysis of this amplifier has been covered in Chapter 6. In that analysis we have calculated voltage gain, input impedance, output impedance, and current gain. These values are reproduced below.

$$\text{From Equ.(6-84), } A_v \approx \frac{h_{fb}R'_L}{h_{ib}} = \frac{h_{fb}}{h_{ib}} \left(\frac{R_C R_L}{R_C + R_L} \right)$$

$$\text{From Equ.(6-89), } Z_i = R_E \parallel Z_e = R_E \parallel h_{ib}$$

$$\text{From Equ.(6-91), } Z_o = r_o \parallel R_C$$

11.10.1 Low Frequency Response

As discussed in Section 11-16, the lower cut-off frequency of this amplifier due to the input coupling capacitor (C_{ic}) will be,

$$\text{Using Equ.(11-22), } f_{Lic} = \frac{1}{2\pi(r_s + Z_i)C_{ic}}$$

$$\text{From Equ.(6-89), } f_{Lic} = \frac{1}{2\pi(r_s + R_E \parallel h_{ib})C_{ic}} \text{ Hz} \quad (11-75)$$

The lower cut-off frequency of due to the output coupling capacitor (C_{oc}) will be,

$$\text{Using Equ.(11-32), } f_{Loc} = \frac{1}{2\pi(Z_o + R_L)C_{oc}}$$

$$\text{Using Equ.(6-91), } f_{Loc} = \frac{1}{2\pi(r_o \parallel R_C + R_L)C_{oc}} \text{ Hz} \quad (11-76)$$

The amplifier that uses bypass capacitor (like C_B of Fig. 11-35) will have another lower cut-off frequency due to this bypass capacitor. To calculate this cut-off frequency, we have to calculate the

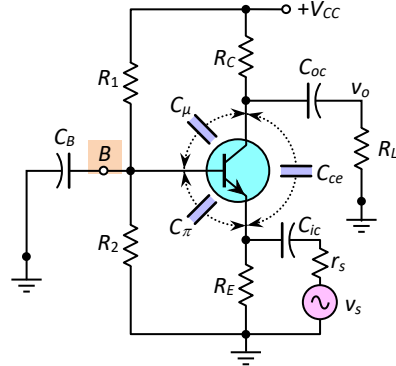


Fig.11-35: CB amplifier with voltage divider bias

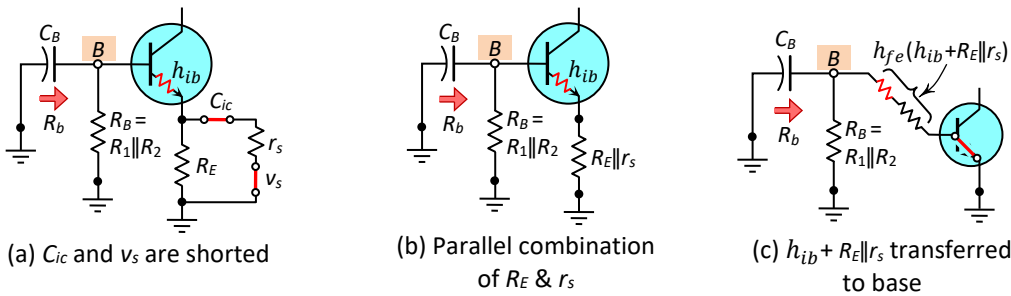


Fig.11-36: Resistance seen from base

resistance seen by C_B capacitor at point B shown in Fig.11-36. $R_E \parallel r_s$ will come in series with the base-to-emitter resistance (h_{ib}) which is shown in Fig.11-36(b). That is total resistance at the emitter side will be $h_{ib} + R_E \parallel r_s$. When this resistance is transferred to the base, it will be multiplied by h_{fe} and come in parallel with R_B . Therefore, the of total resistance seen at point B will be,

$$R_b = h_{fe}(h_{ib} + R_E \parallel r_s) \parallel R_B \quad (11-77)$$

Where, $R_B = R_1 \parallel R_2$. The cut-off frequency will occur when the reactance of C_B equals to R_b . That is,

$$X_{C_B} = R_b$$

Using Equ.(11-77),

$$X_{C_B} = h_{fe}(h_{ib} + R_E \parallel r_s) \parallel R_B$$

or,

$$\frac{1}{2\pi f_{LB} C_B} = h_{fe}(h_{ib} + R_E \parallel r_s) \parallel R_B$$

$\therefore$

$$f_{LB} = \frac{1}{2\pi[h_{fe}(h_{ib} + R_E \parallel r_s) \parallel R_B]C_E} \quad (11-78)$$

Among the three cut-off frequencies, the overall lower cut-off frequency (f_L) of the amplifier has to be determined as described in Section 11.6.2. If the frequencies are far apart f_L will be the smallest of the three. If two of them are equal, f_L will be 1.55 times either one. If all are equal, then f_L will be double on any value. However, generally in CB amplifier the cut-off frequency is set by the **bypass capacitor C_{ic}** .

Example 11-11

For the common base amplifier of Fig.11-37, calculate the lower cut-off frequency. Assume, $r_o = 200$ k.

Solution:

To determine r_e we need the value of I_E . Here also, we have to use exact analysis of the voltage divider biasing circuit.

Using Equ.(5-55),

$$R_{Th} = \frac{R_1 R_2}{R_1 + R_2} = \frac{40.1 \text{ k} \times 18 \text{ k}}{40.1 \text{ k} + 18 \text{ k}} \approx 12.4 \text{ k} = R_B$$

$$\text{Using Equ.(5-54),} \quad E_{Th} = \frac{V_{CC} R_2}{R_1 + R_2} = \frac{12 \text{ V} \times 18 \text{ k}}{40.1 \text{ k} + 18 \text{ k}} \approx 3.72 \text{ V}$$

$$\text{Using Equ.(5-57),} \quad I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (h_{fe} + 1)R_E} = \frac{3.72 \text{ V} - 0.7 \text{ V}}{12.4 \text{ k} + (100 + 1)1.5 \text{ k}} \approx 18.43 \text{ } \mu\text{A}$$

$$\text{Collector current} \quad I_C = h_{fe} I_B = 18.43 \text{ } \mu\text{A} \times 100 = 1.843 \text{ mA} \approx I_E$$

$$\therefore \quad r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{1.843 \text{ mA}} = 14.1 \text{ } \Omega \approx h_{ib}$$

$$\text{Using Equ.(11-75),} \quad f_{Lic} = \frac{1}{2\pi(0.6 \text{ k} + 1.5 \text{ k} \parallel 0.014 \text{ k}) \times 1.5 \text{ } \mu\text{F}} \approx 173 \text{ Hz}$$

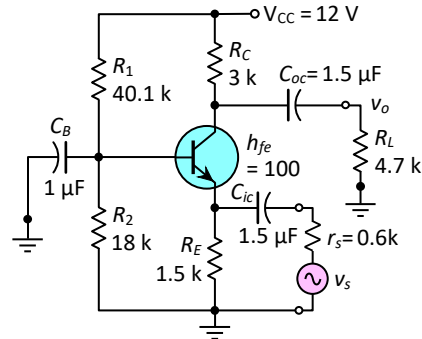


Fig.11-37: CE amplifier for Example 11-11

Using Equ.(11-76),
$$f_{Loc} = \frac{1}{2\pi(200 \text{ k} \parallel 3 \text{ k} + 4.7 \text{ k}) \times 1.5 \text{ } \mu\text{F}} \approx 14 \text{ Hz}$$

Using Equ.(11-78),
$$f_{LB} = \frac{1}{2\pi[100(0.6 \text{ k} \parallel 1.5 \text{ k} + 0.014 \text{ k}) \parallel 12.4 \text{ k}] \times 1 \text{ } \mu\text{F}} \approx 17 \text{ Hz}$$

Among these three lower cut-off frequencies, 173 Hz is approximately 1 decade larger than the other two. So, the lower cut-off frequency of this amplifier will be $f_L = 173 \text{ Hz}$ [Ans.]

Comments: It is described in Chapter 11 that for CB amplifier the lower cut-off frequency is set by the input coupling capacitor, C_{ic} .

11.10.2 High Frequency Response

By using the hybrid- π model of transistor, we can determine the high frequency response of amplifiers. The hybrid- π models of BJT have been discussed in Chapter 6. By incorporating the parasitic capacitances the high frequency equivalent circuit of the CB amplifier can be drawn as shown in Fig.11-38.

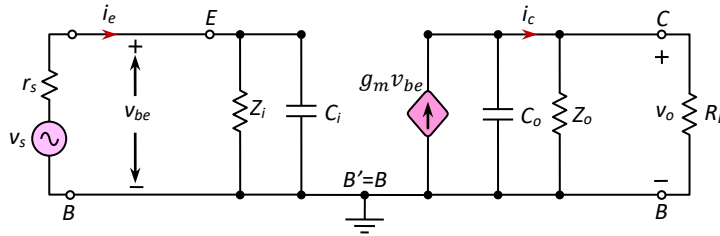


Fig.11-38: High frequency equivalent circuit of CB amplifier

In CB amplifier there is no Miller effect as the voltage gain is positive. Moreover the value of the emitter-collector capacitance (C_{ce}) is very small. Therefore, the high frequency input capacitance will be the sum of base-emitter junction capacitance, C_{π} , and input wiring capacitance, C_{wi} . That is,

$$C_i = C_{\pi} + C_{wi} \quad (11-79)$$

Similarly, high frequency output capacitance will be the sum of base-collector junction capacitance, C_{μ} , and the output wiring capacitance, C_{wo} . That is,

$$C_o = C_{\mu} + C_{wo} \quad (11-80)$$

We have already derived the higher cut-off frequency for the input and output capacitance. The higher cut-off frequency due to the input capacitance is given by Equ.(11-39),

$$f_{Hi} = \frac{1}{2\pi(r_s \parallel Z_i)C_i} \text{ Hz} \quad (11-39)$$

Here, the value of input impedance will be,

$$Z_i = R_E \parallel Z_e = R_E \parallel h_{ib} \quad (11-81)$$

Using Equ.(11-81) into Equ.(11-39) we get,

∴

$$f_{Hi} = \frac{1}{2\pi(r_s \parallel R_E \parallel h_{ib})C_i} \text{ Hz} \quad (11-82)$$

Similarly the higher cut-off frequency due to the output capacitance is given by Equ.(11-47),

$$f_{Ho} = \frac{1}{2\pi(Z_o \parallel R_L)C_o} \text{ Hz} \quad (11-47)$$

From Fig.11-38, the value of output impedance will be,

$$Z_o = r_o \parallel R_C \quad (11-83)$$

∴

$$f_{Ho} = \frac{1}{2\pi(r_o \parallel R_C \parallel R_L)C_o} \text{ Hz} \quad (11-84)$$

Here also, the overall higher cut-off frequency (f_H) of the amplifier has to be determined as described in Section . If the frequencies are far apart f_H will be the smaller one. If they are equal, f_H will be 1.55 times either one.

Example 11-12

Determine the upper cut-off frequency of the amplifier of Example 11-11. Assume that the transistor used in the amplifier has $C_{be} = 10$ pF, $C_{bc} = 5$ pF, $C_{ce} = 2$ pF, $r_o = 200$ k, and 1 pF wiring capacitance on each side.

Solution:

Here, $C_{be} = C_\pi = 10$ pF, $C_{bc} = C_\mu = 5$ pF and in Example 10-, we have found $h_{ib} = 14.1 \Omega$

Using Equ.(11-79), $C_i = C_\pi + C_{wi} = 10$ pF + 1 pF = 11 pF

Using Equ.(11-80), $C_o = C_\mu + C_{wo} = 5$ pF + 1 pF = 6 pF

Using Equ.(11-82), $f_{Hi} = \frac{1}{2\pi \times (0.6 \text{ k} \parallel 1.5 \text{ k} \parallel 0.014 \text{ k}) \times 11 \text{ pF}} \text{ Hz} \approx 1.07 \text{ GHz}$

Using Equ.(11-84), $f_{Ho} = \frac{1}{2\pi \times (200 \text{ k} \parallel 31 \text{ k} \parallel 4.7 \text{ k}) \times 6 \text{ pF}} \text{ Hz} \approx 6.63 \text{ MHz}$

Among these two frequencies, the higher cut-off frequency will be $f_H = f_{Ho} = 6.63 \text{ MHz}$ [Ans.]

Comments: In CB amplifier generally, the output parasitic capacitance (C_o) determines the higher cut-off frequency, due to the larger value of effective R .

Comparing with the result of Example 11-9, we find that the higher cut-off frequency of CB amplifier is greater than that of CE amplifier. This is due to the absence of the Miller Effect. So, we can say that the common-base amplifier has higher bandwidth compared to the CE amplifier.

11.10.3 α -cut-off frequency

The value of current gain (α) of a transistor in CB configuration is approximately 1. Like the β , the value of α also depends on frequency: decreases with the increase in frequency. The frequency at

which the value of α falls to $\alpha/\sqrt{2} \approx 0.707$ is called the **alpha cut-off frequency** and is denoted as f_α . Generally the value of f_α is a little greater than f_T . The variation of α with frequency and f_α is shown in Fig.11-39.

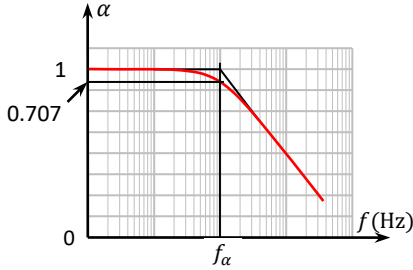


Fig.11-40: Variation of α with frequency and the position of f_α

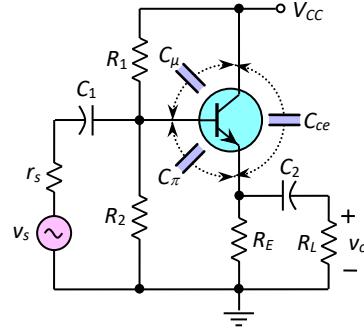


Fig.11-39: CC amplifier with voltage divider bias

11.11 Frequency Response of Common Collector Amplifier

The circuit diagram of a common base amplifier with voltage divider bias is given in Fig.11-40 with both the external and internal capacitances. The low frequency and high frequency response of this circuit is described below.

As described in Chapter 6, the voltage gain, input impedance, output impedance, and current gain of this amplifier are,

From Equ.(6-119),
$$A_v = \frac{h_{fc} R_E}{h_{ic} + h_{fc} R_E} \approx 1$$

From Equ.(6-121),
$$Z_i = R_B \parallel Z_b = R_B \parallel (h_{ic} + h_{fc} R'_L)$$

From Equ.(6-123),
$$Z_o = Z_e \parallel R_E = \left(\frac{r_s \parallel R_B + h_{ic}}{h_{fc}} \right) \parallel R_E \text{ [neglecting } r_o]$$

11.11.1 Low Frequency Response

Already we have studied how find low frequency response in Section 6.12.1. We can use the expression derived in that section. The lower cut-off frequency of this amplifier due to the input coupling capacitor will be,

From Equ.(11-49),
$$f_{Lic} = \frac{1}{2\pi(r_s + Z_i)C_{ic}}$$

Putting the value of Z_i

From Equ.(6-121),
$$f_{Li} = \frac{1}{2\pi[r_s + R_B \parallel (h_{ic} + h_{fc} R'_L)]C_{ic}} \text{ Hz} \quad (11-85)$$

The lower cut-off frequency of due to the output coupling capacitor will be,

From Equ.(11-50),

$$f_{Loc} = \frac{1}{2\pi(Z_o + R_L)C_{oc}}$$

Using Equ.(6-123),

$$f_{Loc} = \frac{1}{2\pi[(r_s \parallel R_B + h_{ic})/h_{fc} \parallel R_E + R_L]C_{oc}} \text{ Hz} \quad (11-86)$$

Unlike the other two amplifiers, CC amplifier generally do not need bypass capacitor. So this amplifier will have only two lower cut-off frequencies.

Among these cut-off frequencies, the overall lower cut-off frequency (f_L) of the amplifier has to be determined as before. If the frequencies are far apart f_L will be the smallest of the two. If they are equal, f_L will be 1.55 times either one.

Example 11-13

For the common collector amplifier of Fig.11-41, calculate the lower cut-off frequency. Assume, $r_o = 200 \text{ k}$.

Solution:

To determine r_e we need the value of I_E . Here also, we have to use exact analysis.

Using Equ.(5-55),

$$R_{Th} = \frac{R_1 R_2}{R_1 + R_2} = \frac{13 \text{ k} \times 16 \text{ k}}{13 \text{ k} + 16 \text{ k}} \approx 7.17 \text{ k} = R_B$$

Using Equ.(5-54),

$$E_{Th} = \frac{V_{CC} R_2}{R_1 + R_2} = \frac{12 \text{ V} \times 16 \text{ k}}{16 \text{ k} + 13 \text{ k}} \approx 6.62 \text{ V}$$

Using Equ.(5-57),

$$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (h_{fe} + 1)R_E} = \frac{6.62 \text{ V} - 0.7 \text{ V}}{7.17 \text{ k} + (100 + 1)1 \text{ k}} \approx 54.73 \text{ } \mu\text{A}$$

Collector current

$$I_C = h_{fe} I_B = 54.73 \text{ } \mu\text{A} \times 100 = 5.47 \text{ mA} \approx I_E$$

$$\therefore r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{5.47 \text{ mA}} = 4.75 \text{ } \Omega$$

$$\therefore h_{ie} \approx h_{fe} r_e = 100 \times 4.75 \text{ } \Omega = 475 \text{ } \Omega = h_{ic}$$

$$R'_L = R_E \parallel R_L = 1 \text{ k} \parallel 100 \text{ } \Omega \approx 90.91 \text{ } \Omega$$

and

$$h_{fc} \approx h_{fe} = 100$$

Using Equ.(11-85),

$$f_{Lic} = \frac{1}{2\pi[0.6 \text{ k} + 7.17 \text{ k} \parallel (0.475 \text{ k} + 100 \times .091)] \times 1.6 \text{ } \mu\text{F}} \approx 21.16 \text{ Hz}$$

Using Equ.(11-86),

$$f_{Loc} = \frac{1}{2\pi[(0.6 \text{ k} \parallel 7.17 \text{ k} + 0.475 \text{ k})/100 \parallel 1 \text{ k} + 0.1 \text{ k}]15 \text{ } \mu\text{F}} \approx 96 \text{ Hz}$$

The overall lower cut-off frequency of the amplifier will be $f_L \approx 96 \text{ Hz}$.

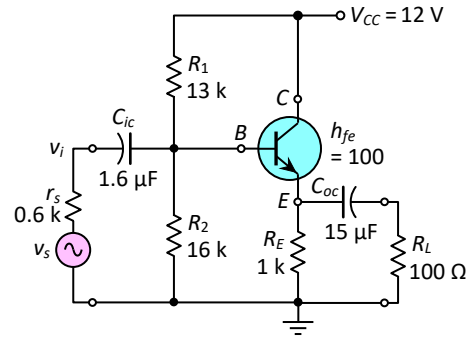


Fig.11-41: CC amplifier designed in Example 11-13

Comments: In Chapter 10, we have learnt that for CC amplifier the lower cut-off frequency is set by the output coupling capacitor. Here also we found the same.

11.11.2 High Frequency Response

By using the hybrid- π model of transistor, we can determine the high frequency response of amplifiers. The hybrid- π models of BJT have been discussed in Chapter 6. Using that hybrid- π model, we can draw the equivalent circuit of CC amplifier as shown in Fig.11-42.

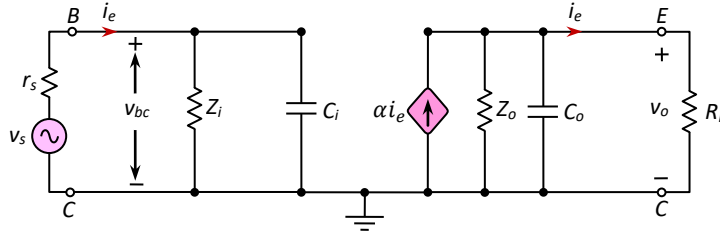


Fig.11-42: High frequency equivalent circuit of CC amplifier

In CC amplifier, there is no Miller effect as the voltage gain is approximately +1. Therefore, high frequency input capacitance will be the sum of base emitter junction capacitance, C_{μ} , and wiring capacitance, C_{wi} . That is,

$$C_i = C_{\mu} + C_{wi} \quad (11-87)$$

Similarly, high frequency output capacitance will be the sum of base collector junction capacitance, C_{ce} , and wiring capacitance, C_{wo} . That is,

$$C_o = C_{ce} + C_{wo} \quad (11-88)$$

We have already derived the higher cut-off frequency for the input and output capacitance. The higher cut-off frequency due to the input capacitance is given by Equ.(11-39),

$$f_{Hi} = \frac{1}{2\pi C_i (r_s \parallel Z_i)} \text{ Hz} \quad (11-39)$$

From Equ.(6-121) we get,

$$Z_i = R_B \parallel r_{b'c} \approx R_B [\because r_{b'c} = 1/h_{oe} \text{ is very large}]$$

$$\therefore \boxed{f_{Hi} = \frac{1}{2\pi [r_s \parallel R_B] (h_{ic} + h_{fc} R_L') C_i} \text{ Hz}} \quad (11-89)$$

Similarly the higher cut-off frequency due to the output capacitance is given by Equ.(11-47),

$$f_{H(C_o)} = \frac{1}{2\pi (Z_o \parallel R_L) C_o} \text{ Hz} \quad (11-47)$$

Here, the value of output impedance will be,

From Equ.(6-123),	$Z_o = Z_e \parallel R_E = \left(\frac{r_s \parallel R_B + h_{ic}}{h_{fc}} \right) \parallel R_E$ [neleccting r_o]	Chapter 5 has to be formatted again
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Putting the value of Z_o into Equ.(11-47)

	$f_{Ho} = \frac{1}{2\pi[(r_s \parallel R_B + h_{ic})/h_{fc}] \parallel R_E \parallel R_L] C_o} \text{ Hz}$	(11-90)
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Here also, the overall higher cut-off frequency (f_H) of the amplifier has to be determined as described in Section . If the frequencies are far apart f_L will be the smaller one. If they are equal, f_H will be 1.55 times either one.

Example 11-14

Determine the upper cut-off frequency of the amplifier of Example . Assume that the transistor used in the amplifier has $C_{be} = 10$ pF, $C_{bc} = 5$ pF, $C_{ce} = 2$ pF, $r_o = 200$ k, and 1 pF wiring capacitance on each side.

Solution:

Here, $C_{be} = C_\pi = 10$ pF, $C_{bc} = C_\mu = 5$ pF and $C_{ce} = 2$ pF. In Example 10-, we have found $h_{fc} = 100$, $h_{ic} = 475 \Omega$, $R'_L = 90.91 \Omega$, and $R_B = 7.17$ k.

Using Equ.(11-87), $C_i = C_\mu + C_{wi} = 5 \text{ pF} + 1 \text{ pF} = 6 \text{ pF}$

Using Equ.(11-88), $C_o = C_{ce} + C_{wo} = 2 \text{ pF} + 1 \text{ pF} = 3 \text{ pF}$

Using Equ.(11-89), $f_{Hi} = \frac{1}{2\pi[0.6 \text{ k} \parallel 7.17 \text{ k} \parallel (0.475 \text{ k} + 100 \times 0.091 \text{ k})]6 \text{ pF}} \approx 50 \text{ MHz}$

Using Equ.(11-90), $f_{Ho} = \frac{1}{2\pi[\{(0.6 \text{ k} \parallel 7.17 \text{ k} + 0.475 \text{ k})/100\} \parallel 1 \text{ k} \parallel 1 \text{ k}\}3 \text{ pF}} \approx 5.7 \text{ GHz}$

Between these two frequencies, the higher cut-off frequency will be $f_H = 50 \text{ MHz}$

Comments: In CC amplifier generally, the input parasitic capacitance determines the higher cut-off frequency, due to the larger value of R and C .

11.12 Frequency Response of CS FET Amplifier

The circuit diagram of a common source (CS) single-stage amplifier is given in Fig.11-. In this circuit the inter-electrode capacitances are also shown. Here we will describe the low and high frequency response of this amplifier.

The analysis of this amplifier has been covered in Chapter 9. In that analysis we have calculated the voltage gain, input impedance, output impedance, and current gain. The values are given below.

From Equ.(9-4), $Z_i \approx R_G = R_1 \parallel R_2$ (9-4)

From Equ.(9-5), $Z_o = R_D \parallel r_d$ (9-5)

From Equ.(9-12) $A_v \approx -g_m(R_D \parallel R_L) = -g_m R'_L$ (9-12)

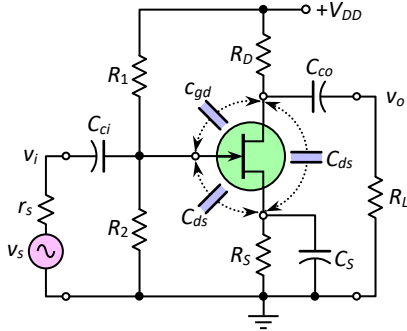


Fig.11-44: Single stage CS FET amplifier with external and internal capacitances

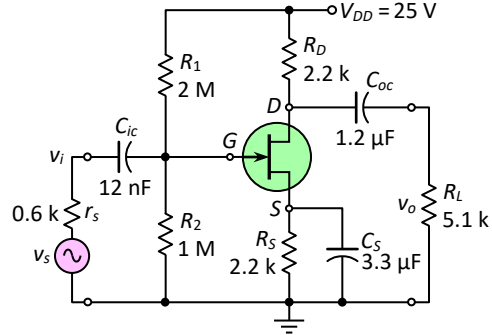


Fig.11-44: Single stage CS FET amplifier for Example 11-15

11.12.1 Low Frequency Response

As discussed in Section , the lower cut-off frequency of this amplifier due to the input coupling capacitor will be,

Using Equ.(11-22),
$$f_{Lic} = \frac{1}{2\pi(r_s + Z_i)C_{ic}}$$

Using Equ.(9-4),

$$f_{Lic} = \frac{1}{2\pi(r_s + R_1 \parallel R_2)C_{ic}} \text{ Hz} \quad (11-91)$$

The lower cut-off frequency due to the output coupling capacitor will be,

Using Equ.(11-32),
$$f_{Loc} = \frac{1}{2\pi(Z_o + R_L)C_{oc}}$$

Using Equ.(9-5),

$$f_{Loc} = \frac{1}{2\pi(r_d \parallel R_D + R_L)C_{oc}} \text{ Hz} \quad (11-92)$$

Another lower cut-off frequency will be produced for the bypass capacitor, C_S . To calculate this cut-off frequency, we have to calculate the resistance seen by C_S capacitor at the source terminal. In [section 9.6.1](#) we have calculated the value of C_S based on the resistance seen at the source point. From that discussion, we find that the resistance at the source terminal is $1/g_m$ which is seen by C_S . The cut-off frequency will occur when the reactance of C_S will be equal to R_e . That is,

$$X_{C_S} = \frac{1}{g_m} \text{ [at } f = f_{LS}]$$

or,

$$\frac{1}{2\pi f_{LS} C_S} = \frac{1}{g_m}$$

∴

$$f_{LS} = \frac{g_m}{2\pi C_S} \quad (11-93)$$

If the lower cut-off frequencies $[f_{Lic}, f_{Loc}, f_{LS}]$ are far apart, the overall lower cut-off frequency of the amplifier, f_L , will be the smallest one of them. If two of them are equal and much larger than the third one (by at least one decade), f_L will be 1.55 times either one. If all are equal, then f_L will be double of any value. However, generally in CS amplifier the lower cut-off frequency is set by the bypass capacitor C_S . It is further illustrated in Example 11-15.

Example 11-15

Fig.11-44 shows a common-source (CS) single-stage FET amplifier. Calculate the lower cut-off frequency of this amplifier. Assume $g_m = 3.8 \text{ mS}$, $r_d = \infty$

Using Equ.(11-91),
$$f_{Lic} = \frac{1}{2\pi(0.6 \text{ k} + 2 \text{ M} \parallel 1 \text{ M}) \times 12 \text{ nF}} \approx 20 \text{ Hz}$$

Using Equ.(11-92),
$$f_{Loc} = \frac{1}{2\pi(2.2 \text{ k} + 5.1 \text{ k}) \times 1.2 \text{ }\mu\text{F}} \approx 18 \text{ Hz}$$

Using Equ.(11-93),
$$f_{LS} = \frac{g_m}{2\pi C_S} = \frac{3.8 \text{ mS}}{2\pi \times 3.3 \text{ }\mu\text{F}} \approx 183 \text{ Hz}$$

As, 183 Hz is much (≈ 1 decade) larger than the other two, the lower cut-off frequency of the amplifier will be $f_L = 183 \text{ Hz}$ [Ans.].

Comments: In Chapter 10, we have found that for CS amplifier the source bypass capacitor (C_S) determines the lower cut-off frequency.

11.12.2 High Frequency Response

By using the hybrid- π model of transistor, we can determine the high frequency response of amplifiers. The hybrid- π models of JFET have been discussed in Chapter 9. Using that hybrid- π model we can draw the equivalent circuit of CS amplifier as shown in Fig.11-45.

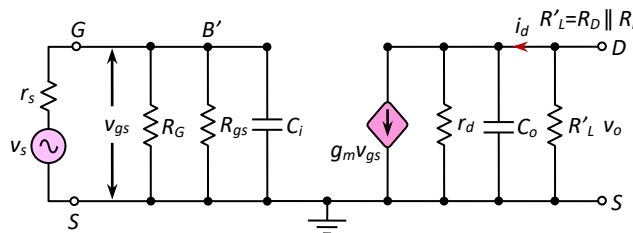


Fig.11-45: High frequency equivalent circuit of CS amplifier

Here, C_i is the total high frequency capacitance in the input side. It includes the gate-source junction capacitance (C_{gs}), Miller capacitance (C_{Mi}) and the input wiring capacitance of the circuit (C_{wi}). The voltage gain of the amplifier is $-g_m r_d \parallel R'_L$ and $R'_L = R_D \parallel R_L$. Therefore, the value of the Miller capacitance will be,

$$C_{Mi} = C_\mu(1 + g_m r_d \parallel R'_L) \quad (11-94)$$

and,
$$C_i = C_{gs} + C_{Mi} + C_{wi} = C_{gs} + C_{gd}(1 + g_m r_d \| R'_L) + C_{wi} \quad (11-95)$$

Similarly, C_o is the total high frequency capacitance at output side. It includes the drain-source interelectrode capacitance (C_{ds}), Miller capacitance (C_{Mo}) and the output wiring capacitance of the circuit (C_{wo}). The Miller capacitance will be,

$$C_{Mo} = C_{gd}(1 + g_m r_d \| R'_L) / (g_m r_o \| R'_L) \approx C_{gd} \quad (11-96)$$

$$C_o = C_{ds} + C_{Mo} + C_{wo} = C_{ds} + C_{gd} + C_{wo} \quad (11-97)$$

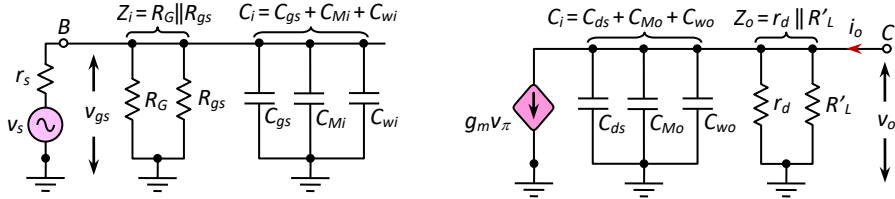


Fig.11-46: All input and output parasitic capacitances of CS amplifier

For better understanding, all the effective resistances and capacitance at high frequencies, the components are shown in Fig.11-46.

We have already derived the higher cut-off frequencies for the input and output capacitance in topic 11.7. The higher cut-off frequency due to the input capacitance is given by Equ.(11-39),

Using Equ.(11-39),
$$f_{Hi} = \frac{1}{2\pi C_i (r_s \| Z_i)} \text{ Hz}$$

Here, the value of input impedance will be,

$$Z_i = R_G \| R_{gs} \approx R_G \quad (11-98)$$

$\therefore$
$$f_{Hi} = \frac{1}{2\pi C_i (r_s \| R_G)} \text{ Hz} \quad (11-99)$$

Similarly the higher cut-off frequency due to the output capacitance is given by Equ.(11-47),

$$f_{Ho} = \frac{1}{2\pi C_o (Z_o \| R_L)} \text{ Hz}$$

Here, the value of output impedance will be,

$$Z_o = r_o \| R'_L = r_o \| R_C \| R_L \quad (11-100)$$

$\therefore$
$$f_{Ho} = \frac{1}{2\pi C_o (r_d \| R_D \| R_L)} \text{ Hz} \quad (11-101)$$

Here also, the overall higher cut-off frequency (f_H) of the amplifier has to be determined as described in Section . If the frequencies are far apart, then f_H will be the smaller one. If they are equal, then f_H will be 1.55 times either one.

Example 11-16

For the CS amplifier given in Fig.11-15, calculate the higher cut-off frequency and bandwidth. Assume, $C_{gd} = 4$ pF, $C_{gs} = 10$ pF, $C_{ds} = 1$ pF and 1 pF wiring capacitance on each side. Also assume $g_m = 3.8$ mS, $r_d = \infty$.

Solution:

$$R'_L = R_D \parallel R_L = 2.2 \text{ k} \parallel 5.1 \text{ k} \approx 1.54 \text{ k}$$

Using Equ.(11-95), $C_i = 10 \text{ pF} + 4 \text{ pF}(1 + 3.8 \text{ mS} \times 1.54 \text{ k}) + 1 \text{ pF} \approx 38.4 \text{ pF}$

Using Equ.(11-97), $C_o = 1 \text{ pF} + 4 \text{ pF} + 1 \text{ pF} = 6 \text{ pF}$

Using Equ.(11-99), $f_{Hi} = \frac{1}{2\pi(0.6 \text{ k} \parallel 2 \text{ M} \parallel 1 \text{ M})38.4 \text{ pF}} \approx 7 \text{ MHz}$

Using Equ.(11-101), $f_{Ho} = \frac{1}{2\pi(2.2 \text{ k} \parallel 5.1 \text{ k})6 \text{ pF}} \approx 17 \text{ MHz}$

Between these two frequencies, the higher cut-off frequency will be $f_H \approx 7 \text{ MHz}$ [Ans.]

The lower cut-off frequency has been calculated in Example 11-15 and found, $f_L = 183 \text{ Hz}$. So the bandwidth of this amplifier will be,

$$BW = f_H - f_L = 7 \text{ MHz} - 183 \text{ Hz} \approx 7 \text{ MHz} \text{ [Ans.]}$$

Comments: Frequency response of CS amplifier is better than CE amplifier.

11.13 Frequency Response of Common Gate Amplifier

The circuit diagram of a common gate amplifier with voltage divider bias is given in Fig.11-47. In this circuit the inter-electrode capacitances are also shown.

AC analysis of this amplifier has been covered in Chapter 9. In that analysis we have calculated voltage gain, input impedance, output impedance, and current gain. These values are reproduced below.

From Equ.(9-65), $A_v \approx g_m R'_L$ (9-65)

From Equ.(9-60), $Z_i \approx R_S \parallel \frac{1}{g_m}$ (9-60)

From Equ.(9-69), $Z_o = r_d \parallel R_D$ (9-69)

Here also, the lower cut-off frequency due to the input coupling capacitor can be calculated using,

Using Equ.(11-22), $f_{Lic} = \frac{1}{2\pi(r_s + Z_i)C_{ic}}$

Using Equ.(9-60),

$$f_{Lic} = \frac{1}{2\pi(r_s + R_s \parallel 1/g_m)C_{ic}} \text{ Hz} \quad (11-102)$$

The lower cut-off frequency due to the output coupling capacitor will be,

From Equ.(11-32),

$$f_{Loc} = \frac{1}{2\pi(Z_o + R_L)C_{oc}}$$

Using Equ.(9-69),

$$f_{Loc} = \frac{1}{2\pi(r_d \parallel R_D + R_L)C_{oc}} \text{ Hz} \quad (11-103)$$

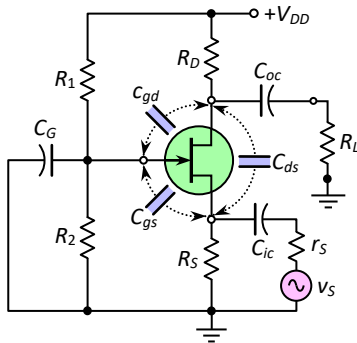


Fig.11-48: Single stage CG FET amplifier with external and internal capacitances

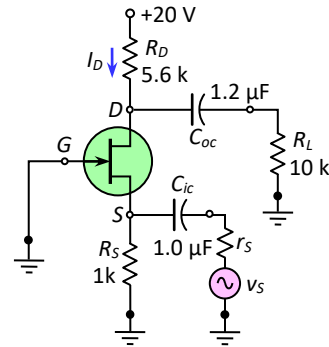


Fig.11-48: Single stage CG FET amplifier for Example 11-17

Another lower cut-off frequency will be produced for the bypass capacitor, C_G . To calculate this cut-off frequency, we have to calculate the resistance seen by C_G capacitor at the source terminal. As the device input resistance R_{gs} is very large, the effective resistance will be only the biasing resistance $R_G = R_1 \parallel R_2$. Therefore, the lower cut-off frequency for this capacitor will be,

$$\therefore f_{LG} = \frac{1}{2\pi R_G C_G} \quad (11-104)$$

Among the three cut-off frequencies, the overall lower cut-off frequency (f_L) of the amplifier has to be determined as described in Section . If the frequencies are far apart f_L will be the smallest of the three. If two of them are equal, f_L will be 1.55 times either one. If all are equal, then f_L will be double on any value. However, generally in CG amplifier the cut-off frequency is set by the **input coupling capacitor C_{ic}** .

Example 11-17

Fig.11-48 shows a common-gate (CG) single-stage FET amplifier with self-bias. Calculate the lower cut-off frequency of this amplifier. Assume $g_m = 3.8 \text{ mS}$, $r_d = 100 \text{ k}\Omega$.

Solution:

The DC analysis of this circuit has been done in Example 9-6 and following results are found:

$g_m = 2 \text{ mS}$. As the value of r_s is not given, we will assume $r_s = 0$.

Using Equ.(11-102),
$$f_{Lic} = \frac{1}{2\pi(0 + 1 \text{ k} \parallel (1/2 \text{ mS}) \times 1 \text{ } \mu\text{F})} \approx \text{Hz}$$

Using Equ.(11-103),
$$f_{Loc} = \frac{1}{2\pi(100 \text{ k} \parallel 5.6 \text{ k} + 10 \text{ k}) \times 1.2 \text{ } \mu\text{F}} \approx \text{Hz}$$

Comments: The higher cut-off frequency of common-gate amplifier is larger than that of common-source amplifier.

11.13.1 High Frequency Response

By using the parasitic capacitances the high frequency equivalent circuit of a CG amplifier is drawn in Fig.11-49.

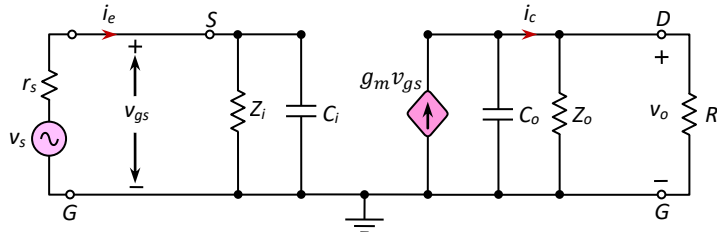


Fig.11-49: AC equivalent circuit of common-gate amplifier

In CG amplifier there is no Miller effect as the voltage gain is positive. Moreover, the value of the drain-source capacitance (C_{ds}) is very small. Therefore, the high frequency input capacitance will be the sum of C_{gs} , and wiring capacitance, C_{wi} . That is,

$$C_i = C_{gs} + C_{wi} \quad (11-105)$$

Similarly, high frequency output capacitance will be the sum of C_{ds} , and wiring capacitance, C_{wo} . That is,

$$C_o = C_{ds} + C_{wo} \quad (11-106)$$

We have already derived the higher cut-off frequency for the input and output capacitance. The higher cut-off frequency due to the input capacitance is given by Equ.(11-39),

$$f_{Hi} = \frac{1}{2\pi C_i (r_s \parallel Z_i)} \text{Hz}$$

Here, the value of input impedance will be,

$$Z_i = R_s \parallel \frac{1}{g_m}$$

$$\therefore \boxed{f_{Hi} = \frac{1}{2\pi [r_s \parallel R_s] (1/g_m) C_i} \text{Hz}} \quad (11-107)$$

Similarly the higher cut-off frequency due to the output capacitance is given by Equ.(11-47),

$$f_{Ho} = \frac{1}{2\pi C_o (Z_o \parallel R_L)} \text{ Hz}$$

Here, the value of output impedance will be,

$$Z_o = r_d \parallel R_D$$

$$\therefore \boxed{f_{Ho} = \frac{1}{2\pi (r_d \parallel R_D \parallel R_L) C_o} \text{ Hz}} \quad (11-108)$$

Here also, the overall higher cut-off frequency (f_H) of the amplifier has to be determined as described in Section . If the frequencies are far apart f_H will be the smaller one. If they are equal, f_H will be 1.55 times of either one.

Example 11-18

Fig.11-50 shows a common-gate FET amplifier. Calculate the lower cut-off frequency and upper cut-off frequency and the bandwidth. $g_m = 3.5 \text{ mS}$, $r_d = \infty$, $C_{gd} = 4 \text{ pF}$, $C_{gs} = 10 \text{ pF}$, $C_{ds} = 1 \text{ pF}$ 1 pF wiring capacitance on each side.

Solution:

Using Equ.(11-102),

$$f_{Lic} = \frac{1}{2\pi (0.6 \text{ k} + 1.6 \text{ k} \parallel 1/3.5 \text{ mS}) 2 \mu\text{F}} \approx 95 \text{ Hz}$$

Using Equ.(11-103),

$$f_{Loc} = \frac{1}{2\pi (2.7 \text{ k} + 5.1 \text{ k}) 3 \mu\text{F}} \approx 6.8 \text{ Hz}$$

$$\text{Using Equ.(11-103), } f_{LG} = \frac{1}{2\pi R_G C_G} = \frac{1}{2\pi (2.84 \text{ M} \parallel 1 \text{ M}) 22 \text{ nF}} \approx 10 \text{ Hz}$$

Among these there cut-off frequencies, f_{Lic} is almost 1 decade higher than the other two. Therefore, the lower cut-off frequency of the amplifier will be $f_L = f_{Lic} = 95 \text{ Hz}$.

Higher cut-off frequency:

$C_{gd} = 4 \text{ pF}$, $C_{gs} = 10 \text{ pF}$, $C_{ds} = 1 \text{ pF}$ 1 pF wiring capacitance on each side.

$$\text{Using Equ.(11-105), } C_i = C_{gs} + C_{wi} = 10 \text{ pF} + 1 \text{ pF} = 11 \text{ pF}$$

$$\text{Using Equ.(11-106), } C_o = C_{ds} + C_{wo} = 1 \text{ pF} + 1 \text{ pF} = 2 \text{ pF}$$

$$\text{Using Equ.(11-107), } f_{Hi} = \frac{1}{2\pi [0.6 \text{ k} \parallel 1.6 \text{ k} \parallel (1/3.5 \text{ mS})] 11 \text{ pF}} \approx 83.8 \text{ MHz}$$

$$\text{Using Equ.(11-108), } f_{Ho} = \frac{1}{2\pi [2.7 \text{ k} \parallel 5.1 \text{ k}] 2 \text{ pF}} \approx 45.1 \text{ MHz}$$

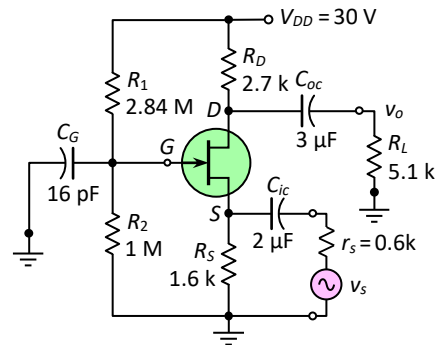


Fig.11-50: CG amplifier designed in Example 11.18

From these two higher cut-off frequencies, we can approximately consider the smaller one as the effective higher cut-off frequency, that is $f_H = f_{Ho} = 45.1 \text{ HHZ}$ [Ans.].

However, we find the exact cut-off frequency from the following normalized gain expression. The exact higher cut-off frequency is the value of f at which the gain falls to -3 dB of the following voltage gain expression considering both the cut-off frequencies,

$$A_v(\text{normalized}) = 20 \log_{10} \left[\frac{1}{\sqrt{1 + (f/83.8 \text{ MHz})^2} \times \sqrt{1 + (f/45.1 \text{ MHz})^2}} \right] \text{ dB}$$

Solving the above equation the higher cut-off frequency is found $f_H \approx 37 \text{ MHz}$ [process is not shown here]

Bandwidth of the amplifier will be: $BW = f_H - f_L = 45.1 \text{ MHz} - 95 \text{ Hz} \approx 45.1 \text{ MHz}$ [Ans.]

Comments: As the two higher cut-off frequencies are close together (not one decade apart), the actual higher cut-off frequency is a bit different.

11.14 Frequency Response of Common Drain Amplifier

The circuit diagram of a common base amplifier with voltage divider bias is given in Fig.11-. As described in Chapter 9, the voltage gain, input impedance, output impedance, and current gain of this amplifier are,

From Equ.(9-40),
$$A_v = \frac{g_m(R_S \parallel R_L)}{1 + g_m(R_S \parallel R_L)} \approx 1$$

From Equ.(9-44),
$$Z_i \approx R_G \quad (9-44)$$

From Equ.(9-49),
$$Z_o = \frac{1}{g_m} \parallel r_d \parallel R_S$$

The low frequency and high frequency response of this circuit is described below.

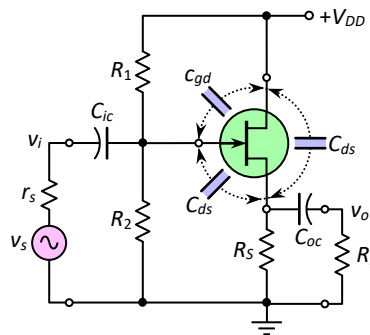


Fig.11-51: Common drain FET amplifier

11.14.1 Low Frequency Response

The lower cut-off frequency due to the input coupling capacitor will be,

Using Equ.(11-22),

$$f_{Lic} = \frac{1}{2\pi(r_s + Z_i)C_{ic}}$$

Using Equ.(9-44),

$$f_{Lic} = \frac{1}{2\pi(r_s + R_G)C_{ic}} \text{ Hz} \quad (11-109)$$

The lower cut-off frequency of due to the output coupling capacitor will be,

Using Equ.(11-32),

$$f_{Loc} = \frac{1}{2\pi(Z_o + R_L)C_{oc}}$$

Using Equ.(9-49),

$$f_{Loc} = \frac{1}{2\pi[(1/g_m)||r_s + R_L]C_{oc}} \text{ Hz} \quad (11-110)$$

Unlike the other two amplifiers, CD amplifier generally do not need bypass capacitor. So this amplifier will have only two lower cut-off frequencies.

Among these cut-off frequencies, the overall lower cut-off frequency (f_L) of the amplifier has to be determined as before. If the frequencies are far apart f_L will be the smallest of the two. If they are equal, f_L will be 1.55 times either one.

11.14.2 High Frequency Response

By using the hybrid- π model of transistor, we can determine the high frequency response of amplifiers. The hybrid- π models of BJT have been discussed in Chapter 9. Using that hybrid- π model of CD transistor we can draw the equivalent circuit of CD amplifier as shown in Fig.11-22.

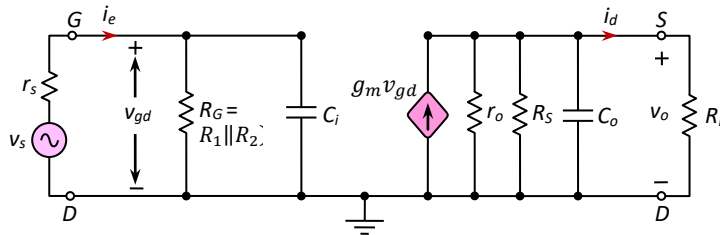


Fig.11-52: High frequency equivalent circuit of a CD amplifier

In CD amplifier there is no Miller effect as the voltage gain is approximately +1. Therefore, high frequency input capacitance will be the sum of C_{gd} , and wiring capacitance, C_{wi} . That is,

$$C_i = C_{gd} + C_{wi} \quad (11-111)$$

Similarly, high frequency output capacitance will be the sum of base collector junction capacitance, C_{ds} , and wiring capacitance, C_{wo} . That is,

$$C_o = C_{ds} + C_{wo} \quad (11-112)$$

We have already derived the higher cut-off frequency for the input and output capacitance. The higher cut-off frequency due to the input capacitance is given by Equ.(11-39),

$$f_{Hi} = \frac{1}{2\pi C_i (r_s \parallel Z_i)} \text{ Hz} \quad (11-113)$$

Putting the value of Z_i from Equ.(9-44),

$$\therefore f_{Hi} = \frac{1}{2\pi (r_s \parallel R_G) C_i} \text{ Hz} \quad (11-114)$$

Similarly the higher cut-off frequency due to the output capacitance is given by Equ.(11-47),

$$f_{Ho} = \frac{1}{2\pi (Z_o \parallel R_L) C_o} \text{ Hz}$$

Here, the value of output impedance will be, $Z_o = r_o \parallel R_s$

Putting this value of Z_o into the above equation,

$$\therefore f_{Ho} = \frac{1}{2\pi [r_o \parallel R_s \parallel R_L] C_o} \text{ Hz} \quad (11-115)$$

Here also, the overall higher cut-off frequency (f_H) of the amplifier has to be determined as described in Section. If the frequencies are far apart f_L will be the smaller one. If they are equal, f_H will be 1.55 times either one.

Example 11-19

A common-drain FET amplifier is given in Fig.11-53. Calculate the lower and the higher cut-off frequencies and the bandwidth. Assume, $g_m = 5 \text{ mS}$, $C_{gd} = 4 \text{ pF}$, $C_{gs} = 10 \text{ pF}$, $C_{ds} = 1 \text{ pF}$ 1 pF wiring capacitance on each side.

Solution:

Calculation of Lower Cut-off Frequency:

Here, $R_G = R_1 \parallel R_2 = 1 \text{ M} \parallel 1.5 \text{ M} = 0.6 \text{ M}$

$$\text{Using Equ.}(11-109), \quad f_{Lic} = \frac{1}{2\pi (0.6 \text{ k} + 0.6 \text{ M}) 27 \text{ nF}} \approx 10 \text{ Hz}$$

$$\text{Using Equ.}(11-110), \quad f_{Loc} = \frac{1}{2\pi [(1/5 \text{ mS}) \parallel 1.6 \text{ k} + 4.3 \text{ k}] 0.25 \text{ }\mu\text{F}} \approx 142 \text{ Hz}$$

Calculation of Higher Cut-off Frequency:

Here, $C_i = C_{gd} + C_{wi} = 4 \text{ pF} + 1 \text{ pF} = 5 \text{ pF}$

and $C_o = C_{ds} + C_{wo} = 1 \text{ pF} + 1 \text{ pF} = 2 \text{ pF}$

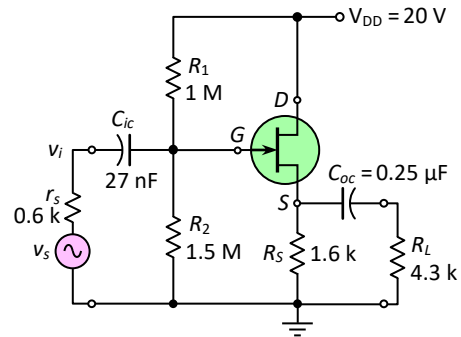


Fig.11-53: CD amplifier for Example 11-19

Using Equ.(11-114),
$$f_{Hi} = \frac{1}{2\pi[0.6 \text{ k}\|0.6 \text{ M}] 5 \text{ pF}} \approx 53 \text{ MHz}$$

Using Equ.(11-115),
$$f_{Ho} = \frac{1}{2\pi[(1/5 \text{ mS})\|1.6 \text{ k}\|4.3 \text{ k}] 2 \text{ pF}} \approx 466 \text{ MHz}$$

The effective higher cut-off frequency will be $f_H = 53 \text{ MHz}$.

Comments: f_{Hi} is smaller than f_{Ho} due to larger effective resistance.

11.15 Frequency Response from Poles and Zeros

The terms Poles and Zeros are mainly used in Control Engineering. **Poles** of a transfer function are the frequencies for which the value of the denominator becomes zero or the value of the transfer function becomes infinity. On the other hand, **zeros** of a transfer function are the frequencies for which the value of the numerator becomes zero or the value of the transfer function becomes zero. Generally in determining the poles and zeroes, the transfer function of a system is represented in frequency domain, not in time domain. To do that a complex frequency, $s = \sigma + j\omega$, is considered. Here, σ is the real part and ω is the imaginary part. For example, let $s = a + jb$. This complex frequency can be represented in a polar coordinate system as in Fig.11-54. Here, the magnitude is $|s| = \sqrt{a^2 + b^2}$, and the angle will be $\phi = \tan^{-1}(b/a)$.

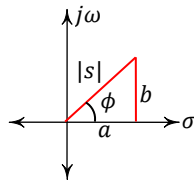


Fig.11-55: Presentation of a complex frequency $s = a + jb$ in a polar coordinate system

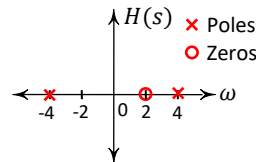


Fig.11-55: Position of poles and zeroes in graph

In the frequency domain, if the only **sine** and **cosine** waves are considered, then s will have only the imaginary part, i.e., $s = j\omega$. Now, let's come back to our topic and consider the following transfer function,

$$H(s) = \frac{K(s-4)}{(s-2)(s+4)}$$

here, K is the gain of the system

From this transfer function we find one zero and two poles. The zero is 4 and the values of the poles are 2 and -4 . The positions of these pole and zeroes is shown in Fig.11-57.

The values of the poles and the zeros of a system determine whether the system is stable, and how well the system performs.

Generally, we can define the transfer function of a system as the ratio of two polynomials, as

$$H(s) = \frac{N(s)}{D(s)} = \frac{K(s + z_1)(s + z_2) \cdots (s + z_m)}{(s + p_1)(s + p_2) \cdots (s + p_n)}$$

$$H(s) = \frac{K(1 + s/z_1)(1 + s/z_2) \cdots (1 + s/z_m)}{(1 + s/p_1)(1 + s/p_2) \cdots (1 + s/p_n)}$$

Where $N(s)$ and $D(s)$ are simple polynomials.

Zeros are the roots of $N(s)$ (the numerator of the transfer function) obtained by setting $N(s) = 0$ and solving for s . Poles are the roots of $D(s)$ (the denominator of the transfer function), obtained by setting $D(s) = 0$ and solving for s . Using Bode's Rule, we can determine the frequency response of the system.

Bode's Rules:

- i) As ω passes a pole frequency the slope of $|H|$ (the magnitude of the transfer function) decreases by 20 dB/dec.
- ii) As ω passes a zero frequency the slope of $|H|$ (the magnitude of the transfer function) increases by 20 dB/dec.

The process is illustrated in Example 11-20.

Example 11-20

Calculate the transfer function and plot the frequency response of the RC circuit given in Fig.11-56(a).

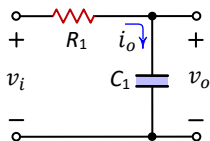


Fig.11-57: An RC circuit for Example 11-20

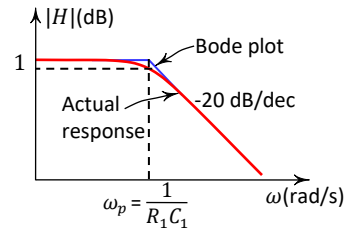


Fig.11-57: Frequency response of the RC circuit for Example 11-20

$$v_o = \frac{v_i(1/j\omega C_1)}{R_1 + (1/j\omega C_1)} = \frac{v_i(1/sC_1)}{R_1 + (1/sC_1)} = \frac{v_i}{sR_1C_1 + 1}$$

$$H(s) = \frac{v_o}{v_i} = \frac{1}{1 + sR_1C_1} = \frac{1}{1 + s/(1/R_1C_1)}$$

From this expression of transfer function we find that $K = 1$, that is, the gain of the circuit is 1. There is no zero but only one pole at $\omega = 1/RC$. Therefore, we can plot the frequency response of the RC circuit starting with a gain of 1 with zero slope and at $\omega = 1/R_1C_1$, the slope will be -20 dB/dec. The plot is given in Fig.11-56(b).

Comments: Actually the circuit is a low-pass filter and its higher cut-off frequency is $\omega_p = 1/R_1C_1$.

11.15.1 Finding Poles by Inspection

Without any serious analysis, we can find the poles in a circuit just by inspections. We can assume that each node in a signal path contributes a pole. We can calculate the pole as below:

- i) Find the total resistance from the node to the AC ground.
- ii) Find the total capacitance from that node to the AC ground.

If the total resistance is R and the total capacitance is C , then we have to add a pole $\omega_p = 1/RC$, in the transfer function if the signal passes through this point. Similarly, we have to decrease the slope of $|H|$ by 20 dB, at ω_p in the frequency response. The process is further illustrated in Example 11-21.

Example 11-21

Fig.11- shows a common-source FET amplifier with stray capacitances. From this figure find out the transfer characteristics and draw the Bode plot.

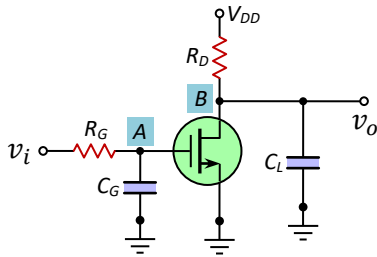


Fig.11-59: A simple CS amplifier for Example 11-21

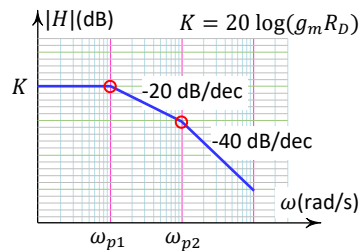


Fig.11-59: Frequency response of the RC circuit for Example 11-21

Solution:

From this figure, we find a pole at node A. If $v_i = 0$, the resistance R_G will be short to ground. C_G is connected to ground from this node. So the value of pole will be, $\omega_{p1} = 1/R_G C_G$. The second pole is found at node B. In AC equivalent circuit, R_D will be connected to ground from this node. Similarly, C_L is connected to ground from this node. Thus, the value of this pole will be, $\omega_{p2} = 1/R_D C_L$. We know that the gain of a CS FET amplifier the voltage gain is $-g_m R_D$. From these poles and voltage gain, we can easily find the transfer function as,

$$H(s) = \frac{v_o}{v_i} = \frac{-g_m R_D}{(1 + s/\omega_{p1})(1 + s/\omega_{p2})}$$

$$H(s) = \frac{-g_m R_D}{(1 + s R_G C_G)(1 + s R_D C_L)}$$

The Bode plot of this transfer function is given in Fig.11-.

Comments: As we represent the slope in dB/dec, the absolute gain has to be converted into dB scale.

11.15.2 Finding Transfer Function from Bode Plot (Frequency Response) Theory

If the Bode magnitude plot of a system is given, we can find the transfer function of the system. The general expression of a transfer function of a system is,

$$H(s) = \frac{N(s)}{D(s)} = \frac{K(s + z_1)(s + z_2) \cdots (s + z_m)}{(s + p_1)(s + p_2) \cdots (s + p_n)}$$

$$H(s) = \frac{K(1 + s/z_1)(1 + s/z_2) \cdots (1 + s/z_m)}{(1 + s/p_1)(1 + s/p_2) \cdots (1 + s/p_n)}$$

From the given Bode plot, we have to find out if there is any slope change in it. The frequency where the plot changes its slope is called **corner frequency**. At each corner frequency, we have to determine the rate of change of slope. According to Bode's Rule, if the slope increases by 20 dB/dec at any frequency (say $\omega = z_1$), we have to add a zero ($1 + s/z_1$) at the numerator. Similarly, if the slope decreases by 20 dB/dec at any frequency (say $\omega = p_1$), we have to add a pole ($1 + s/p_1$) at the denominator. But if the slope increases (or decreases) by 40 dB/dec at any frequency (say $\omega = z_2$), we have to add two zeros ($1 + s/z_2$)² at the numerator (or denominator). The constant K indicates the gain of the system. The value of K can be calculated from the first segment (or the horizontal segment) of the graph. The process is illustrated further in Example 11-, and in Exercise 11-.

Example 11-22

The Bode magnitude plot of an amplifier is shown in Fig.11-. From this response find the transfer function of the amplifier.

Solution:

From the first segment ($\omega = 0.1$ to 1), we can calculate the gain of the amplifier as,

$$20 \log(K) = 20$$

$$\log(K) = 1$$

$$K = \text{antilog}(2) = 10^1 = 10$$

The corner frequencies of this Bode plot are (the frequencies where the slope of the graph is changing):

$$\omega = 1, 10, 60$$

At $\omega = 1$, the slope is increased by $(40 - 20)$ dB/dec = 20 dB/dec. So, a zero exists here and its value will be,

$$(1 + s/z_1) = (1 + s/1)$$

At $\omega = 10$, the slope is decreased by $(0 - 20)$ dB/dec = -20 dB/dec. So, a zero exists here. Its value will be,

$$(1 + s/P_1) = (1 + s/10)$$

Again, at $\omega = 60$, the slope is decreased by 20 dB/dec. So, a pole exists here. Its value will be $(1 + s/P_2) = (1 + s/60)$.

The transfer function will be

$$H(s) = \frac{K(1 + s/z_1)}{(1 + s/P_1)(1 + s/P_2)} = \frac{10(1 + s/1)}{(1 + s/10)(1 + s/60)}$$

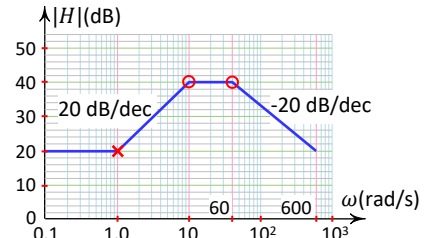


Fig.11-60: Bode plot (magnitude) for Example 11-22

$$H(s) = \frac{10(1 + s/1)}{(1/10)(10 + s)(1/60)(60 + s)}$$

$$H(s) = \frac{6000(1 + s)}{(10 + s)(60 + s)}$$

11.16 Sample Questions

1. What is frequency response, and mid-band frequencies? What do you mean by low frequency response and high frequency response?
2. Define the unit Bel and decibel (dB). Prove that dB for voltage is just double of dB for power.
3. How to use dB unit to calculate the gain of cascaded amplifiers?
4. What are dBm unit and dBV unit? Why we need them?
5. Why do we use logarithmic plot? Explain with figures.
6. What is semi-log and log-log graph paper? Explain decades and octaves in log graph papers.
7. What is frequency response and phase response? Explain with figures.
8. What is Bode plot? Draw a Bode plot (magnitude) and explain different features of this plot.
9. What is Bode plot? Draw a Bode plot (phase) and explain some important features of this plot.
10. Derive the expression for the lower cut-off frequency due to the input coupling capacitor (C_{ic}) of an amplifier.
11. Derive the expression for the phase angle of the input voltage due to the input coupling capacitor (C_{ic}).
12. Draw the Bode magnitude response for the output coupling capacitor (C_{oc}) and explain in log-log graph.
13. Derive the expression for the lower cut-off frequency due to the output coupling capacitor (C_{oc}) of an amplifier.
14. Derive the expression for the phase angle of the output voltage due to the output coupling capacitor (C_{oc}).
15. Draw the Bode magnitude response for the output coupling capacitor (C_{oc}) and explain in log-log graph.
16. Draw the frequency response of an amplifier if the lower cut-off frequencies for (C_{ic}), and (C_{oc}) are far apart from each other and explain.
17. Draw the frequency response of an amplifier if the lower cut-off frequencies for (C_{ic}), and (C_{oc}) have the same value.
18. Derive the expression for the upper cut-off frequency of an amplifier due to the input shunt capacitance (C_i).
19. Derive the expression for the phase angle of input voltage of an amplifier due to the input shunt capacitance (C_i).

20. Derive the expression for the upper cut-off frequency of an amplifier due to the output shunt capacitance (C_o).
21. Derive the expression for the phase angle of output voltage of an amplifier due to the output shunt capacitance (C_o).
22. Draw the frequency response of an amplifier with input shunt capacitance (C_i) and explain.
23. Draw the frequency response of an amplifier with output shunt capacitance (C_o) and explain.
24. What is bandwidth and gain-bandwidth product? Explain with figures.
25. Derive the expression for the lower cut-off frequency due to the input coupling capacitor (C_{ic}) of a common-emitter amplifier.
26. Derive the expression for the lower cut-off frequency due to the output coupling capacitor (C_{oc}) of a common-emitter amplifier.
27. Derive the expression for the lower cut-off frequency due to the emitter bypass capacitor (C_E) of a common-emitter amplifier.
28. Derive the expression for the higher cut-off frequency due to the input shunt capacitance of a common-emitter amplifier.
29. Derive the expression for the lower cut-off frequency due to the output shunt capacitance of a common-emitter amplifier.
30. Derive the expression for the gain-bandwidth product of a common-emitter amplifier.
31. What is beta cut-off frequency (f_β)? Derive the expression for the beta cut-off frequency.
32. What is unity current gain frequency (f_T)? Derive the expression for this frequency.
33. What is the relation between f_T and f_β ?
34. Derive the expression for the lower cut-off frequency due to the input coupling capacitor (C_{ic}) of a common-base amplifier.
35. Derive the expression for the lower cut-off frequency due to the output coupling capacitor (C_{oc}) of a common-base amplifier.
36. Derive the expression for the lower cut-off frequency due to the base bypass capacitor (C_E) of a common-base amplifier.
37. Derive the expression for the higher cut-off frequency due to the input shunt capacitance of a common-base amplifier.
38. Derive the expression for the lower cut-off frequency due to the output shunt capacitance of a common-base amplifier.
39. Derive the expression for the gain-bandwidth product of a common-base amplifier.
40. What is alpha cut-off frequency (f_α)? Derive the expression for the alpha cut-off frequency.
41. Derive the expression for the lower cut-off frequency due to the input coupling capacitor (C_{ic}) of a common-collector amplifier.
42. Derive the expression for the lower cut-off frequency due to the output coupling capacitor (C_{oc}) of a common-collector amplifier.
43. Derive the expression for the lower cut-off frequency due to the collector bypass capacitor (C_E) of a common-collector amplifier.

44. Derive the expression for the higher cut-off frequency due to the input shunt capacitance of a common-collector amplifier.
45. Derive the expression for the lower cut-off frequency due to the output shunt capacitance of a common-collector amplifier.
46. Derive the expression for the gain-bandwidth product of a common-collector amplifier.
47. Derive the expression for the lower cut-off frequency due to the input coupling capacitor (C_{ic}) of a common-source amplifier.
48. Derive the expression for the lower cut-off frequency due to the output coupling capacitor (C_{oc}) of a common-source amplifier.
49. Derive the expression for the lower cut-off frequency due to the source bypass capacitor (C_E) of a common-source amplifier.
50. Derive the expression for the higher cut-off frequency due to the input shunt capacitance of a common-source amplifier.
51. Derive the expression for the lower cut-off frequency due to the output shunt capacitance of a common-source amplifier.
52. Derive the expression for the gain-bandwidth product of a common-source amplifier.
53. Derive the expression for the lower cut-off frequency due to the input coupling capacitor (C_{ic}) of a common-gate amplifier.
54. Derive the expression for the lower cut-off frequency due to the output coupling capacitor (C_{oc}) of a common-gate amplifier.
55. Derive the expression for the lower cut-off frequency due to the gate bypass capacitor (C_E) of a common-gate amplifier.
56. Derive the expression for the higher cut-off frequency due to the input shunt capacitance of a common-gate amplifier.
57. Derive the expression for the lower cut-off frequency due to the output shunt capacitance of a common-gate amplifier.
58. Derive the expression for the gain-bandwidth product of a common-gate amplifier.
59. Derive the expression for the lower cut-off frequency due to the input coupling capacitor (C_{ic}) of a common-drain amplifier.
60. Derive the expression for the lower cut-off frequency due to the output coupling capacitor (C_{oc}) of a common-drain amplifier.
61. Derive the expression for the lower cut-off frequency due to the drain bypass capacitor (C_E) of a common-drain amplifier.
62. Derive the expression for the higher cut-off frequency due to the input shunt capacitance of a common-drain amplifier.
63. Derive the expression for the lower cut-off frequency due to the output shunt capacitance of a common-drain amplifier.
64. Derive the expression for the gain-bandwidth product of a common-drain amplifier.
65. What are poles and zeroes? Explain how we can find the frequency response of a circuit from poles and zeroes.
66. Explain how we can find the poles and zeroes of a network by inspection.
67. Explain how we can find the transfer function from Bode plot.

11.17 Exercise

1. The input impedance and the output impedance of an amplifier are $R_i = R_o = 1\text{ k}$. If 1 V rms is applied at the input, 16 V rms is developed at the output. Calculate the power gain using Equ.(11-2) and the voltage gain using Equ.(11-5).
2. Two amplifiers are cascaded amplifier to increase the overall gain. The gain of the first amplifier is 23 dB . The input voltage of the first amplifier is $v_i = 5\text{ mV (rms)}$. Calculate the output voltage of this stage. If the voltage at the output of the 2nd amplifier is 5 V (rms) , then calculate the voltage gain of this stage both in absolute value and in dB. Also calculate the overall gain of the whole system.
3. Fig.11-61(a) and Fig.11-61(b) show the frequency response of a typical amplifier in semi-log and log-log graph, respectively. Determine the lower and higher cut-off frequency of the amplifier from these graphs

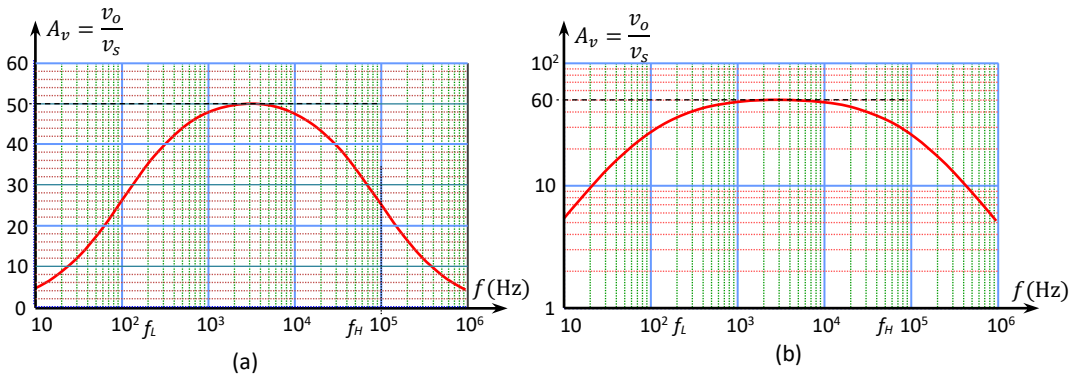


Fig.11-61: Frequency response of an amplifier: (a) Semi-log plot, and (b) Log-log plot

4. The amplifier shown in Fig.11-62 has mid-band gain, $v_o/v_s = 55\text{ dB}$. (i) Calculate the absolute mid-band gain and the gain at lower cut-off frequency both in dB and absolute value. (ii) Lower cut-off frequency, and phase shift of v_i at $f = 1.5\text{ kHz}$.

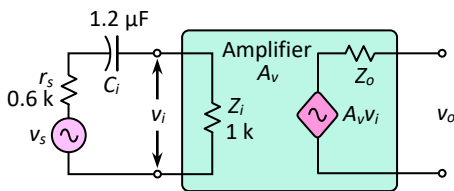


Fig.11-62: Amplifier with input coupling capacitor

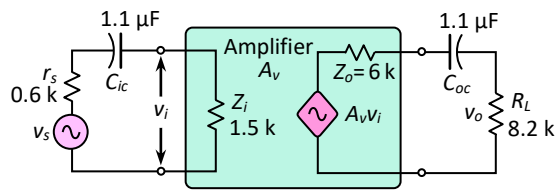


Fig.11-63: Two-port model of an amplifier for Example 11-5

5. The gain of the amplifier of Fig.11-63 is, $A_v = 150$. Calculate the mid-band overall gain (v_o/v_s) of the amplifier, cut-off frequencies due to the input and output coupling capacitors.
6. The open circuit voltage gain of the amplifier of Fig.11-64 is $v_i/v_o = 220$. Calculate the overall mid-band voltage gain and the higher cut-off frequency.

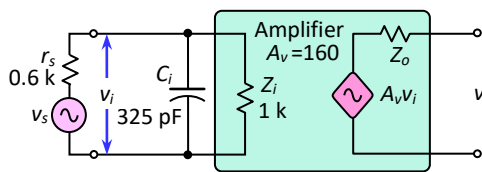


Fig.11-64: Amplifier with input shunt capacitance for Example 11-6

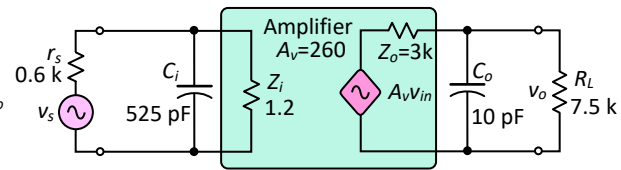


Fig.11-65: Amplifier with input and output shunt capacitance for Example 11-7

7. For the amplifier of Fig.11-65, calculate the overall mid-band gain, cut-off frequencies due the input and output shunt capacitances and plot the frequency response (Bode plot).
8. Determine the lower cut-off frequencies imposed by the coupling and bypass capacitors of the CE amplifier given in Fig.11-66.

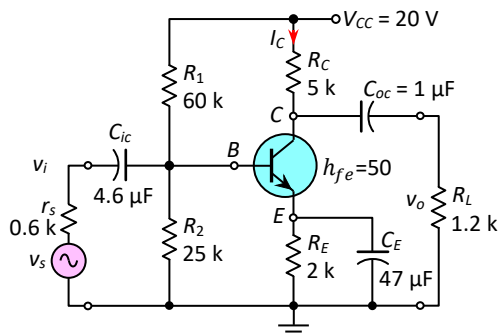


Fig.11-66: CE amplifier for Example 11.8

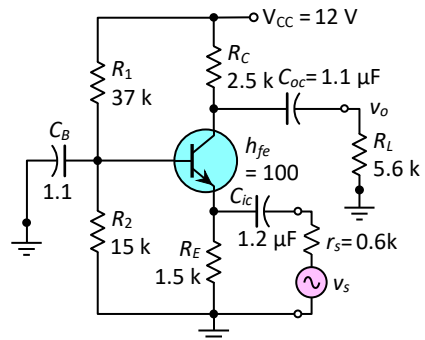


Fig.11-67: CB amplifier for Example 11-11

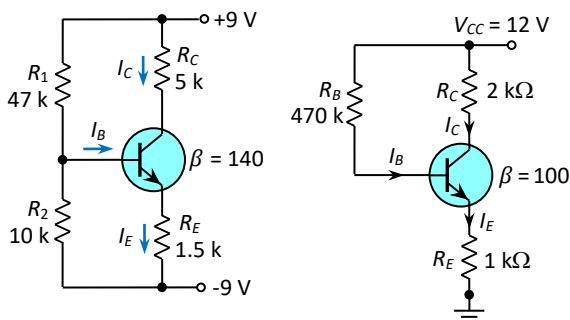


Fig.5-50: Voltage divider bias circuit for Example 5-18

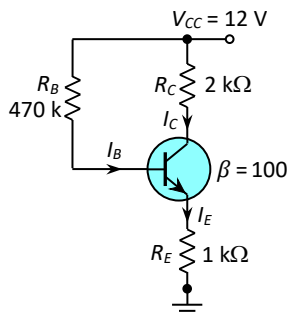


Fig.5-22: Emitter bias for Example 5-7

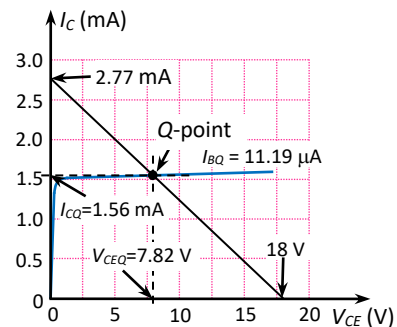


Fig.5-51: Load line and operating point of Example 5-18

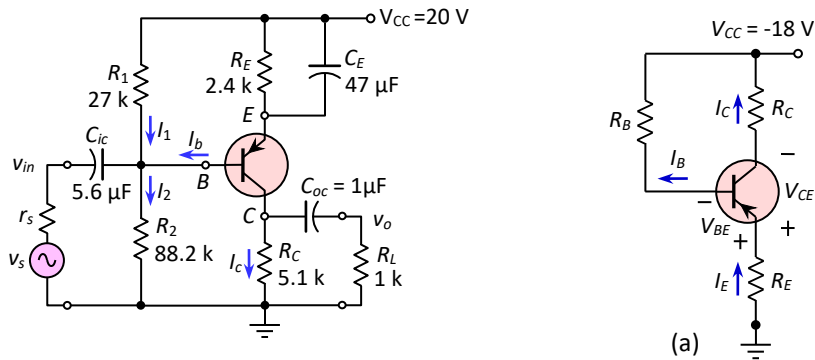


Fig.10-13: CE amplifier designed in Example 9.7

9. Determine the upper cut-off frequency of the amplifier of Exercise 8. Assume that the transistor used in the amplifier has $C_{be} = 12$ pF, $C_{bc} = 4.5$ pF, and $C_{ce} = 3$ pF. $r_o = 250$ k. Neglect wiring capacitances.
10. For the transistor of Exercise 8 & 9 (above two) determine the values of f_β , and f_T .
11. For the common base amplifier of Fig.11-67, calculate the lower cut-off frequency. Assume, $r_o = 250$ k.
12. Determine the upper cut-off frequency of the amplifier of Exercise 11. Assume that the transistor used in the amplifier has $C_{be} = 12$ pF, $C_{bc} = 6.5$ pF, $C_{ce} = 2.5$ pF, $r_o = 300$ k, and 1.2 pF wiring capacitance on each side.
13. For the common base amplifier of Fig.11-67, calculate the lower cut-off frequency. Assume, $r_o = 200$ k.

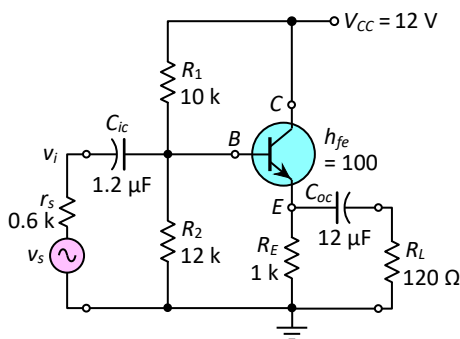


Fig.11-68: CC amplifier designed in Example 11-69

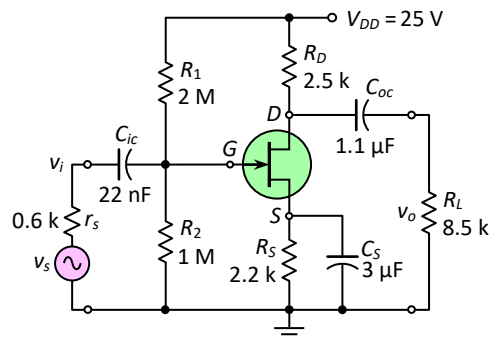


Fig.11-70: Single stage CS FET amplifier for Example 11-15

14. Determine the upper cut-off frequency of the amplifier of Exercise 13. Assume that the transistor used in the amplifier has $C_{be} = 14$ pF, $C_{bc} = 4.7$ pF, $C_{ce} = 3$ pF, $r_o = 260$ k, and 1 pF wiring capacitance on each side.

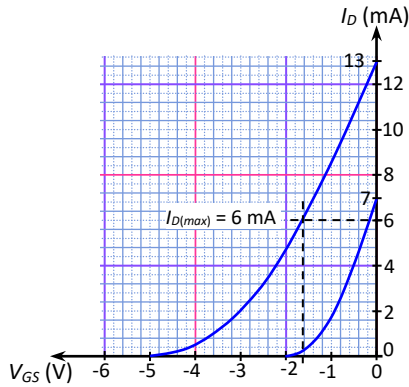


Fig.10-22: Transfer characteristics and bias current for Example 9.10

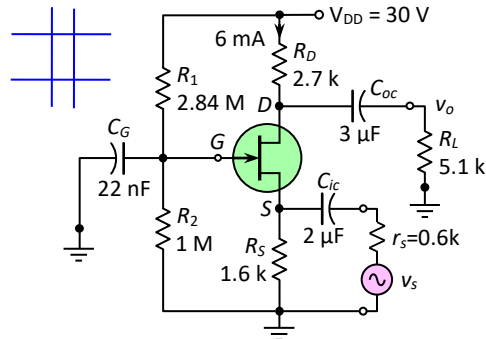


Fig.10-23: CG amplifier designed in Example 9.10

15. Fig.11-70 shows a common-source (CS) single-stage FET amplifier. Calculate the lower cut-off frequency of this amplifier. Assume $g_m = 3.8$ mS, $r_d = \infty$

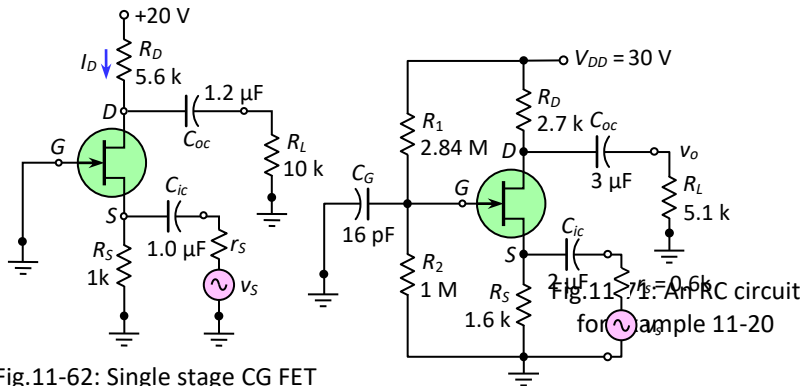


Fig.11-62: Single stage CG FET amplifier for Example 11-17

16. For the CS amplifier given in Fig.11-15, calculate the higher cut-off frequency and bandwidth. Assume, $C_{gd} = 5$ pF, $C_{gs} = 12$ pF, $C_{ds} = 2$ pF, 1 pF wiring capacitance on each side. Also assume $g_m = 4.5$ mS, $r_d = \infty$.
17. Fig.11-48 shows a common-gate (CG) single-stage FET amplifier with self-bias. Calculate the lower cut-off frequency of this amplifier. Assume $g_m = 5.6$ mS, $r_d = 200$ kΩ.
18. Fig.11-50 shows a common-gate FET amplifier. Calculate the lower cut-off frequency and upper cut-off frequency and the bandwidth. $g_m = 4.5$ mS, $r_d = \infty$, $C_{gd} = 4.5$ pF, $C_{gs} = 12$ pF, $C_{ds} = 1.1$ pF and 1 pF wiring capacitance on each side.

19. A common-drain FET amplifier is given in Fig.11-70. Calculate the lower and the higher cut-off frequencies and the bandwidth. Assume, $g_m = 5.5 \text{ mS}$, $C_{gd} = 5.5 \text{ pF}$, $C_{gs} = 8 \text{ pF}$, $C_{ds} = 2.1 \text{ pF}$ 1 pF wiring capacitance on each side.

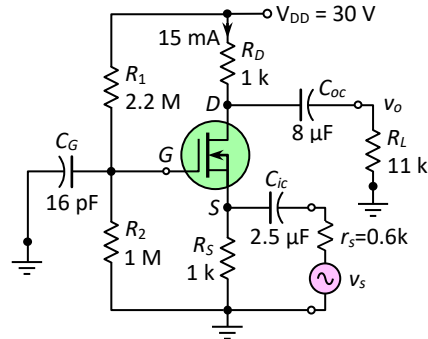
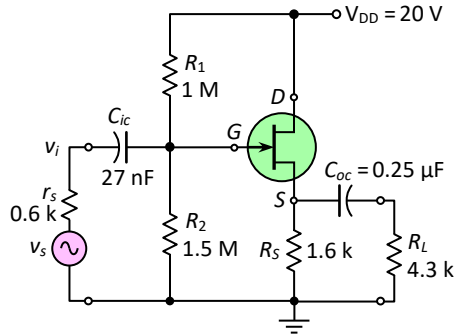


Fig.11-63: CD amplifier for Example 11-19 Fig.10-25: CG amplifier designed in Example 9.11

20. Calculate the transfer function and plot the frequency response of the RC circuit given in Fig.11-71.

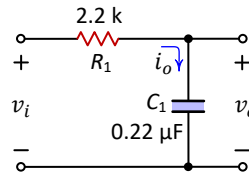


Fig.11-71: An RC circuit for Example 11-20

21. The approximate Bode magnitude plot of a system is given in Fig.11-72. Determine the transfer function of this system. [Ans.: $H(s) = \frac{10^8(s+0.1)^3}{(s+10)^2(s+100)}$]

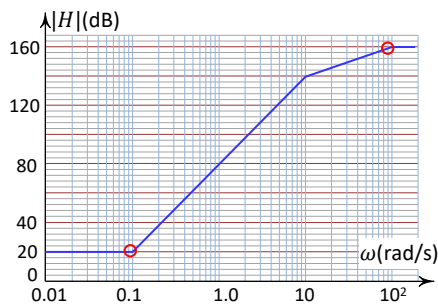


Fig.11-72: Bode magnitude plot for Exercise

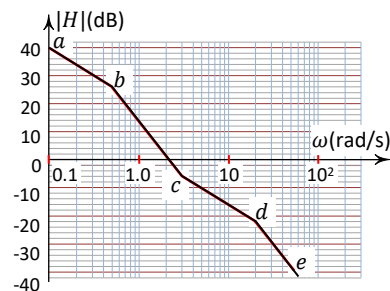


Fig.11-73: Bode magnitude plot for Exercise

22. For the Bode plot of Fig.11-73, determine the slope of different sections of the curve and also determine the pole frequencies. [Ans.]
23. Fig.11-74 shows a common-source FET amplifier with stray capacitances. From this figure find out the transfer characteristics and draw the Bode plot.
24. Fig.11-75 shows a common-emitter BJT amplifier with stray capacitances. Find out the transfer characteristics and draw the Bode plot for this amplifier.

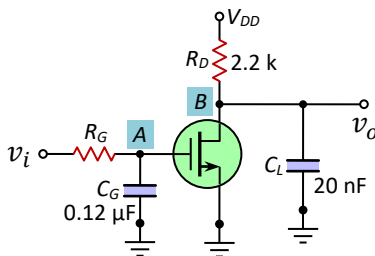


Fig.11-74: A simple CS amplifier for Example 11-21

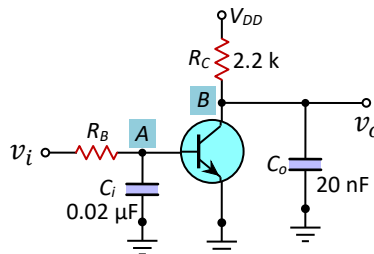


Fig.11-75: A simple CE amplifier for Example 11-21